Pairing of universes of masses +m and -m: exact proofs for dirac16complex and dirac16complex00

The pairing theorems T1, T2 and T3 and the quantum-level reading in the author's primordial gravitational field, with every hypothesis, the verification records and an exact statement of what they do not establish

dirac16complex project (with Claude Opus 5.5)

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Abstract

The author asked on 2026-10-01: "PROVE that Universes of masses {+mass, -mass} are created in pairs for each case of the dirac16complex and the dirac16complex00 fields." This document gives the exact answer computed by the code in `Revision/` for the author's primordial metric. Proved, for both fields and with every hypothesis stated: T1 (chirality pairing): the chirality matrix Γ maps every configuration Ψ of the theory with mass and coupling (m, λ) to the configuration $\Gamma\Psi$ of the theory with $(-m, -\lambda)$, with $\mathcal{L}_{m,\lambda}[\Gamma\Psi] = -\mathcal{L}_{-m,-\lambda}[\Psi]$, solutions to solutions and the energy-momentum tensor and the current reversed, so that the pair has zero total energy-momentum, current and charge as classical bilinears, in every gravitational field. T2 (mirror pairing): Γ combined with a Pin(4,4) reflection of character -1 maps (m, λ) to $(-m, \lambda)$ with $\mathcal{L} \to +\mathcal{L}$; in the author's field it is the mirror across the Z2 brane $z = \pi/2$ (an ASSUMED construction), and the partner has EQUAL, not opposite, energy-momentum. T3 (Kohn-Sham level, dirac16complex): the block map of Γ, with the brane parities exchanged and the tip angle $\theta \to \pi - \theta$, maps every self-consistent instantaneous Kohn-Sham state with (m, λ, θ) onto one with $(-m, +\lambda, \pi - \theta)$, with equal levels, occupations, Kohn-Sham energies and energy-momentum profiles (ASSUMED Z2 brane, mean-field level); a completion of 2026-10-08 adds the 16-component statement S6 (every component of the energy-momentum tensor and of the current is unchanged, S and Q change sign), carries the filling convention onto the $-m$ member and demonstrates T3 numerically with two independent Kohn-Sham solvers (demonstrations, not part of the proof). The quantum-level reading for dirac16complex: the chirality image carries the Krein metric $-B$ and is the same quantum system re-labelled; two independently quantised universes of masses $+m$ and $-m$ have identical one-particle ($\lambda = 0$) spectra (flat space, or frozen coefficients at a point) and their generators do not cancel. A corollary: a T1 pair taken as the complete classical source of the author's metric is a zero source, and the Einstein equations then have no solution for $H > 0$. Not established by any of these equations: a creation process, a rate, a probability or an amplitude for creating universes, in pairs or otherwise. T1, T2, T3 and the quantum reading Q are proved, each under its stated hypotheses; the creation of pairs is not.

1 The request and the exact answer

The author's request, quoted from `Revision/README.md`: "PROVE that Universes of masses {+mass, -mass} are created in pairs for each case of the dirac16complex and the dirac16complex00 fields."

What a universe of mass m means in this record: a configuration of one of the two fields with mass parameter m and coupling λ in the author's primordial gravitational field, i.e. a solution of the Euler-Lagrange equations of section 2.4. Two such universes are paired when an explicit invertible map takes every solution of the theory with mass m to a solution of the theory with mass $-m$ and takes its Lagrangian, energy-momentum tensor and current to stated multiples of those of the partner.

The exact answer, for both fields:

1. Proved (sections 4 to 8): the pairing theorems T1 and T2, the quantum-level reading Q for dirac16complex, the corollary C1 for the field equations of a_4 and the Kohn-Sham-level theorem T3 for dirac16complex, each with its exact hypotheses (T2 and T3 use the ASSUMED Z2 construction at the brane).

2. Not proved, and not derivable from these equations: that such universes are CREATED. The equations are field equations on a fixed gravitational background and their canonical

quantisation; no equation of this record produces a pair of universes from anything, and no creation process, rate or amplitude follows from them (section 11).

3. The Kohn-Sham level T3 of `Revision/SPEC.md` section 9 is proved in section 8 for the instantaneous (adiabatic) mean-field Kohn-Sham states of `Revision/kohn_sham/ks-theory.json`; the time-dependent Kohn-Sham problem is open. Its completion (section 8.5) closes three gaps found by an adversarial verification and adds two numerical demonstrations.

Every formula and number below is taken from the Revision reports listed in section 9; nothing is taken from the earlier stages of the repository. Interpretations are labelled as such.

Related Revision documents. The Kohn-Sham model of dirac16complex in the deflating field, whose states T3 maps, is described in `Revision/docs/KOHN_SHAM_DEFLATING_FIELD` (block reduction, instantaneous ground and first excited states along the prescribed history $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$, Mermin thermodynamics, the full cross-check of the Rust solver against the independent reference, 31 of 31 checks). The Lovelock tensors behind the field equations for a_4 and b_4 used by corollary C1 are described in `Revision/docs/LOVELOCK_GKD`. The field equations for a_4 and b_4 with the Kohn-Sham source (`Revision/field_equations_a4/ks_source`) show that no recorded Kohn-Sham state is an admissible source of the author's metric (section 8.4), so T3 is a statement about test-field states on a prescribed background. The dark-sector hypotheses for both fields are investigated in `Revision/docs/DARK_SECTOR_HYPOTHESES`; that record establishes neither Hypothesis nor Hypothesis00 (it states what each field can and cannot produce under its stated observer assumption), and nothing in it is used by, or changes, the theorems of this document.

2 Setting and conventions

2.1 The author's metric and coordinates

The coordinates are the author's, $x_0, x_1, \ldots, x_7$, in the author's order: x_0 is the hidden space direction, with $z = 6Hx_0 \in (0, \pi/2)$ and $H > 0$ the author's constant, x_1, x_2, x_3 are ordinary 3-space (inflating), x_4 is the time, and x_5, x_6, x_7 are the three extra times, which DEFLATE exponentially as a_4 increases. (Until 2026-10-14 this document named them $x_1, \ldots, x_8$, with the hidden direction x_8 listed last; the user ordered the author's coordinates on 2026-10-14.) The metric of `Revision/SPEC.md` section 1, as changed by the author on 2026-10-14, is

$$\begin{aligned} ds^2 = {} & \cot^2 z\, dx_0^2 + e^{2a_4(x_4)} \sin^{1/3} z\, (dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 \\ & - e^{-2a_4(x_4)}\, b_4(x_4) \sin^{1/3} z\, (dx_5^2 + dx_6^2 + dx_7^2), \end{aligned}$$

that is $g_{55} = g_{66} = g_{77} = -e^{-2a_4(x_4)} b_4(x_4) \sin^{1/3}(6Hx_0)$ with one metric function $b_4(x_4) > 0$ of the time, never 1 ($b_4 = 1$ is the withdrawn metric of 2026-10-01, for which this document was first written; `Revision/metric_b4/METRIC_CHANGE_PROVENANCE.md`); the extra times deflate with the scale factor $e^{-a_4} b_4^{1/2} \sin^{1/6} z$. T1, T2, Q and T3 hold for arbitrary functions $a_4(x_4)$ and $b_4(x_4)$ (the pairing verifiers treat both as arbitrary); the Kohn-Sham record of section 8 uses the prescribed history $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$. The metric has signature (4,4) (space-like x_0, x_1, x_2, x_3; time-like x_4, x_5, x_6, x_7), with $\sqrt{|g|} = b_4^{3/2} \cos z$. The diagonal vielbein is $e^a{}_\mu = f_a \delta^a_\mu$ with $f_0 = \cot z$, $f_1 = f_2 = f_3 = e^{a_4} \sin^{1/6} z$, $f_4 = 1$ and $f_5 = f_6 = f_7 = e^{-a_4} b_4^{1/2} \sin^{1/6} z$; the frame metric is

$\eta = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ in the order $x_0,\dots,x_7$. Frame indices $a = 0,\dots,7$ are aligned with the coordinates, so γ^0 is the hidden-direction gamma $\gamma^{(x_0)}$ and γ^4 the time gamma $\gamma^{(x_4)}$.

The canonical spin connection (vielbein postulate) has the nonzero components $\omega_{x_i,\,i4} = a_4' e^{a_4} \sin^{1/6} z$ and $\omega_{x_i,\,i0} = H e^{a_4} \sin^{1/6} z$ for the inflating directions $i = 1,2,3$, and $\omega_{x_t,\,4t} = -(a_4' - \frac{b_4'}{2b_4}) e^{-a_4} b_4^{1/2} \sin^{1/6} z$ and $\omega_{x_t,\,t0} = -H e^{-a_4} b_4^{1/2} \sin^{1/6} z$ for the deflating extra times $t = 5,6,7$, with $\Omega_{x_0} = \Omega_{x_4} = 0$. Contracted with the gammas, $\gamma^\mu \Omega_\mu = 3H\gamma^0 + \frac{3}{4}\frac{b_4'}{b_4}\gamma^4$: each inflating direction contributes $\frac{a_4'}{2}\gamma^4 + \frac{H}{2}\gamma^0$ and each deflating extra time $(-\frac{a_4'}{2} + \frac{b_4'}{4b_4})\gamma^4 + \frac{H}{2}\gamma^0$, so the a_4' terms of the time direction cancel, the b_4' terms and the hidden-direction terms add.

2.2 Clifford data

The gammas are the author's real 16×16 matrices T16, re-constructed in Revision code from the author's tau formulas, with $\gamma^a = T16[a]$ for $a = 0,\dots,7$ (γ^0 the hidden direction, γ^4 the time). They are real signed permutation matrices with $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$ and $(\gamma^a)^T = \eta_{aa}\gamma^a$. Further:

- $C = \gamma^0\gamma^1\gamma^2\gamma^3$ is real and symmetric, $C^2 = 1$, $C\gamma^a$ is real and antisymmetric and $C\gamma^a C^{-1} = -(\gamma^a)^T$; the Dirac adjoint is $\bar\Psi = \Psi^\dagger C$.
- $\Gamma = \gamma^0\gamma^1\gamma^2\gamma^3\gamma^4\gamma^5\gamma^6\gamma^7 = \mathrm{diag}(-I_8, I_8)$.
- $B = -iC\gamma^4$ is Hermitian with $B^2 = 1$ and signature (8,8).
- $S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$ satisfy the commutation relations of so(4,4).
- The 16-dimensional representation is irreducible under Pin(4,4) (commutant of dimension 1) and splits under Spin(4,4) into two inequivalent irreducible 8-dimensional representations, the chiral halves $\Gamma = -1$ (components 1 to 8, written ψ_-) and $\Gamma = +1$ (components 9 to 16, written ψ_+); every γ^a exchanges the two halves.

2.3 The two fields and the Lagrangian

dirac16complex is Ψ: 16 complex ANTICOMMUTING (Grassmann) components, a Pin(4,4) spinor, quantised canonically (section 6). dirac16complex00 is Φ: 16 complex COMMUTING components that transform as a Pin(4,4) spinor, a classical field. Both are coupled to gravity through the vielbein and the canonical spin connection. The theorems are stated with the letter Ψ for both fields; "both statistics" means that the statement and its proof hold for Grassmann components and for commuting components. With $S = \bar\Psi\Psi$, $D_\mu\Psi = \partial_\mu\Psi + \Omega_\mu\Psi$, $D_\mu\bar\Psi = \partial_\mu\bar\Psi - \bar\Psi\Omega_\mu$, $\Omega_\mu = \frac{1}{2}\omega_{\mu ab}S^{ab}$ and $\gamma^\mu = e^\mu{}_a\gamma^a$, the Lagrangian density of both fields is

$$\mathcal{L}_{m,U}[\Psi] = \sqrt{|g|}\,[K[\Psi] - mS - U(S)], \qquad K[\Psi] = \tfrac{1}{2}(\bar\Psi\gamma^\mu D_\mu\Psi - (D_\mu\bar\Psi)\gamma^\mu\Psi),$$

written $\mathcal{L}_{m,\lambda}$ for $U(S) = \frac{\lambda}{2}S^2$. It is real for both statistics.

2.4 Field equation, energy-momentum tensor, current

The Euler-Lagrange equations of $\mathcal{L}_{m,U}$ are, for both statistics,

$$E_{m,U}[\Psi] \equiv \gamma^\mu D_\mu \Psi - \big(m + U'(S)\big)\Psi = 0, \qquad (D_\mu \bar\Psi)\gamma^\mu = -\big(m + U'(S)\big)\bar\Psi.$$

The symmetric (Belinfante) energy-momentum tensor of the theory record is

$$T^\nu{}_\mu = \delta^\nu_\mu \frac{\mathcal{L}}{\sqrt{|g|}} - \frac{1}{4}\Big(\bar\Psi\gamma^\nu D_\mu\Psi - (D_\mu\bar\Psi)\gamma^\nu\Psi + \bar\Psi\gamma_\mu D^\nu\Psi - (D^\nu\bar\Psi)\gamma_\mu\Psi\Big),$$

with the energy density $\rho = -T^{x_4}{}_{x_4}$ and the pressures $p_\mu = T^\mu{}_\mu$ (no sum) of `Revision/SPEC.md` section 4; for homogeneous on-shell states $\rho = mS + U$ and $p = SU' - U$. The conserved current is $J^\mu = -i\bar\Psi\gamma^\mu\Psi$, with the charge density $J^{x_4} = \Psi^\dagger B\Psi$ and the charge $Q = \int b_4^{3/2}\cos z\,\Psi^\dagger B\Psi\, d^7x$.

Convention note. The pairing records use the tensor

$$T_{\mu\nu} = \tfrac{1}{4}(\bar\Psi\gamma_\mu D_\nu\Psi + \bar\Psi\gamma_\nu D_\mu\Psi - (D_\mu\bar\Psi)\gamma_\nu\Psi - (D_\nu\bar\Psi)\gamma_\mu\Psi) - g_{\mu\nu}\,\mathcal{L}/\sqrt{|g|},$$

which is minus the tensor above, and the currents $\bar\Psi\gamma^\mu\Psi$ (Wolfram) and $i\bar\Psi\gamma^\mu\Psi$ (sympy). Every statement of T1, T2 and Q about T and J is linear and homogeneous in them ($T' = \pm T$, $J' = \pm J$), so it holds unchanged for every overall sign and constant factor.

2.5 The field equation in the author's metric

Written out in the author's metric, the field equation of both fields is

$$\begin{aligned} &e^{-a_4}\sin^{-1/6} z\,(\gamma^1\partial_1 + \gamma^2\partial_2 + \gamma^3\partial_3)\Psi + \gamma^4\partial_4\Psi + e^{a_4}b_4^{-1/2}\sin^{-1/6} z\,(\gamma^5\partial_5 + \gamma^6\partial_6 + \gamma^7\partial_7)\Psi \\ &\quad + \tan z\,\gamma^0\partial_0\Psi + 3H\gamma^0\Psi + \tfrac{3}{4}\tfrac{b_4'}{b_4}\gamma^4\Psi = \big(m + U'(S)\big)\Psi, \end{aligned}$$

where $G = 3H\gamma^0 + \frac{3}{4}(b_4'/b_4)\gamma^4 = \gamma^\mu\Omega_\mu$ is the spin-connection term in the diagonal vielbein of section 2.1 and in the field variables Ψ; in this frame and these variables it is nonzero for every $H > 0$ (its γ^0 coefficient is $3H$), and $G^2 = (9H^2 - \frac{9}{16}(b_4'/b_4)^2)I_{16}$, so $G\Psi \neq 0$ for every $\Psi \neq 0$ wherever $(b_4'/b_4)^2 \neq 16H^2$ (on the history $b_4'/b_4 = \frac{4}{3}H$ of section 2.1, $G^2 = 8H^2$; `nontriviality_1_dirac16complex`, `nontriviality_2_dirac16complex00`). This is non-triviality [1] and [2] of the theory record only in the qualified sense stated there: the value belongs to the frame and the variables. In the frame boosted in the (x_0, x_4) plane with rapidity $6Hx_4 + b_0$ the H part is removed and $\frac{3}{4}(b_4'/b_4)(\cosh b\,\gamma^4 - \sinh b\,\gamma^0)$ remains; no boost in this plane removes the b_4' part (`boosted_frame_gammaOmega_formula`, `boosted_frame_removes_only_the_H_term`); the rescaling $\Psi = b_4^{-3/4}\sin^{-1/2} z\,\chi$ removes the whole term (completely for $U = 0$; for $U \neq 0$ it becomes the x_0- and x_4-dependent coupling $\lambda S[\chi]/(b_4^{3/2}\sin z)$: `rescaling_removes_the_connection_term`, `rescaled_equation_quadratic_potential`); and the canonical spin connection drops out of $\mathcal{L}$ (`connection_free_lagrangian_same_equations`). What is frame-independent is that Ω_μ vanishes in no frame, because its curvature is the Riemann tensor and the metric is curved for every $H > 0$ (`boosted_frame_curvature_nonzero`, `spin_curvature_equals_Riemann`), and that the vielbein

factors enter every derivative term. In the chiral block form $\Psi = (\psi_-, \psi_+)$ every gamma is block off-diagonal, with upper-right block $\bar{\tau}^a$ and lower-left block τ^a, and $\bar{\tau}^0 = \tau^0 = I_8$; the 16 equations become the two 8-component equations

$$\sum_a f_a^{-1} \bar{\tau}^a \partial_a \psi_+ + 3H\psi_+ + \tfrac{3}{4}\tfrac{b_4'}{b_4}\bar{\tau}^4\psi_+ = V\psi_-,$$
$$\sum_a f_a^{-1} \tau^a \partial_a \psi_- + 3H\psi_- + \tfrac{3}{4}\tfrac{b_4'}{b_4}\tau^4\psi_- = V\psi_+, \qquad V = m + U'(S).$$

2.6 Records for section 2

item	Wolfram records	sympy records
metric, vielbein, $\sqrt{\lvert g\rvert} = b_4^{3/2}\cos z$	`metric_is_the_authors`, `sqrt_det_g`, `primordial_vielbein`	`metric_from_vielbein_equals_SPEC`, `sqrt_det_g_equals_b4_cos_z`
spin connection and $\gamma^\mu\Omega_\mu = 3H\gamma^0 + \frac{3}{4}(b_4'/b_4)\gamma^4$	`omega_components`, `Omega_components`, `gammaOmega_total`, `gammaOmega_per_direction`, `primordial_connection_depends_on_b4`	`geometry.spin_connection_components`, `geometry.connection_depends_on_b4`, `gamma_mu_Omega_mu_equals_3H_gamma_x0_plus_b4_term`, `a4_time_terms_cancel_b4_terms_add`
gammas	`coordinate_map`, `Clifford_relation`, `reality`, `signed_permutation_matrices`, `symmetry_pattern`	`clifford_relation`, `reality_signed_permutations`, `symmetry_pattern`
C, Γ, B, S^{ab}	`C_real_symmetric`, `C_squared_identity`, `C_gamma_real_antisymmetric`, `C_gamma_C_inverse`, `Gamma_definition`, `Gamma_diag`, `B_Hermitian`, `B_squared_identity`, `B_signature_8_8`, `S_Lorentz_algebra`	`C_real_symmetric_involution`, `C_gamma_antisymmetric`, `C_conjugation`, `chirality_diag`, `B_hermitian_involution_signature`, `S_lorentz_algebra`
Pin(4,4) and Spin(4,4)	`Pin44_irreducible_commutant_dim_1`, `Spin44_commutant_dim_2_chiral_projectors`, `chiral_halves_irreducible`, `chiral_halves_inequivalent_intertwiners_0`, `Spin_Pin_reflection_swaps_halves`	`pin_commutant_dimension_1`, `spin_commutant_dimension_2`, `spin_halves_irreducible`, `spin_halves_inequivalent`, `reflections_exchange_halves`

item	Wolfram records	sympy records
reality of $\mathcal{L}$	L_real_G, L_real_C	grassmann_lagran gian_real, commutin g_lagrangian_real
Euler-Lagrange equations	EL_Psi_G, EL_Psibar_G, EL_Psi_C, EL_Psibar_C, T1_Euler_Lagrang e_derived_primor dial_grassmann, T1_Euler _Lagrange_derived_primor dial_commuting	grassmann_euler_ lagrange_psi_variation, grassmann_euler_ lagrange_psibar_ variation, commutin g_euler_lagrange _psi_variation, commutin g_euler_lagrange _psibar_variation
energy-momentum tensor	T_symmetric_part _Belinfante_G, T_symmet ric_part_Belinfante_C, kinetic_sum_on_shell_G, kinetic_sum_on_shell_C	grassmann_emt_symmetric, commuting_emt_symmetric, grassmann_homoge neous_on_shell_rho_p, commuting_homoge neous_on_shell_rho_p
current and charge	current_conserva tion_identity_G, current_conserva tion_identity_C, charge_density_i s_Krein_form_G, charge_d ensity_is_Krein_form_C	grassmann_curren t_conservation, commutin g_current_conservation
field equation in the metric, block form, non-triviality	Dirac_operator_e xplicit_G, Dirac_op erator_explicit_C, block_form, nontrivi ality_1_dirac16complex, nontriviality_2_ dirac16complex00	nontriviality_Om ega_zero_iff_flat
scope of non-triviality: frame, variables, frame-independent content	boosted_frame_ga mmaOmega_formula, boosted_frame_re moves_only_the_H_term, boosted_frame_cu rvature_nonzero, rescaling_remove s_the_connection_term, rescaled_equatio n_quadratic_potential, connection_free_ lagrangian_same_ equations, spin_cur vature_equals_Riemann	boosted_frame_ga mmaOmega_formula, boosted_frame_re moves_only_the_H_term, boosted_frame_cu rvature_nonzero, rescaling_remove s_the_connection_term, rescaled_equatio n_quadratic_potential, connection_free_ lagrangian_same_ equations, spinor_c urvature_equals_riemann

3 Algebraic lemmas

All four lemmas are exact matrix identities of the Revision gammas, verified by WolframScript and independently by sympy (records after each proof).

Lemma 1 (the chirality matrix). Γ is real and symmetric, $\Gamma^2 = 1$, $\Gamma\gamma^a = -\gamma^a\Gamma$ for every a, $\Gamma C = C\Gamma$ and $\Gamma S^{ab} = S^{ab}\Gamma$. Hence $\Gamma\Omega_\mu = \Omega_\mu\Gamma$ for EVERY connection $\Omega_\mu = \frac{1}{2}\omega_{\mu ab}S^{ab}$, and $D_\mu(\Gamma\Psi) = \Gamma D_\mu\Psi$, $D_\mu(\bar\Psi\Gamma) = (D_\mu\bar\Psi)\Gamma$.

Proof. Γ is the product of all eight different γ^a. Moving γ^b through it passes the seven factors γ^a, $a \neq b$, which anticommute with γ^b, and the factor γ^b itself, which commutes with it: the sign is $(-1)^7 = -1$. C is a product of four and S^{ab} ($a \neq b$) a multiple of a product of two different gammas, so both commute with Γ. $\Gamma = \mathrm{diag}(-I_8, I_8)$ is computed exactly; it is real, symmetric and squares to 1. Γ is constant and commutes with Ω_μ, which gives the statements about D_μ. QED.

Wolfram records	**sympy records**
`Gamma_properties`, `Gamma_diag`, `Gamma_anticommutes_with_gammas`, `Gamma_commutes_with_C`, `S_commutes_with_Gamma`	`gammas.Gamma`, `gammas.Gamma_anticommutes`, `chirality_anticommutes`, `chirality_C_relation`, `S_preserves_C_and_commutes_with_Gamma`

Lemma 2 (bilinears under the chirality map). For $\Psi' = \Gamma\Psi$: $\bar\Psi' = \bar\Psi\Gamma$; $\bar\Psi' M\Psi' = (-1)^k\bar\Psi M\Psi$ for every product M of k gammas; in particular S is invariant and every bilinear with γ^a, $\gamma^a S^{bc}$ or $S^{bc}\gamma^a$ changes sign. Moreover $\Gamma B\Gamma = -B$.

Proof. By Lemma 1, $\bar\Psi' = \Psi^\dagger\Gamma^\dagger C = \Psi^\dagger\Gamma C = \Psi^\dagger C\Gamma = \bar\Psi\Gamma$. Then $\bar\Psi' M\Psi' = \bar\Psi\,\Gamma M\Gamma\,\Psi$, and $\Gamma M\Gamma = (-1)^k M$ by Lemma 1. Finally $\Gamma B\Gamma = -iC\,\Gamma\gamma^4\Gamma = iC\gamma^4 = -B$. QED.

Wolfram records	**sympy records**
`T1_kernel_scalar`, `T1_kernel_kinetic`, `T1_kernel_connection` (all 512 triples), `T1_kernel_field_equation`, `Q_Krein_metric_of_images`	`T1.general_field.matrix_identities` (456 kinetic and connection basis matrices), `Q.image_krein_metric`

Lemma 3 (reflections and their Pin(4,4) lifts). Let n be a frame direction and R_n the diagonal matrix with -1 in place n and $+1$ elsewhere (the reflection of that direction; $R_n\eta R_n = \eta$).

1. $(\gamma^n)^T C\gamma^n = -\eta_{nn}C$.
2. $\gamma^n\gamma^a(\gamma^n)^{-1} = -(R_n)^a{}_b\gamma^b$ and $\gamma^n S^{ab}(\gamma^n)^{-1} = (R_n)^a{}_c(R_n)^b{}_d S^{cd}$.
3. $P_n := \Gamma\gamma^n$ is plus or minus the product of the seven other gammas, hence an element of Pin(4,4); $P_n^{-1}\gamma^a P_n = (R_n)^a{}_b\gamma^b$ (it covers R_n); $P_n^\dagger C P_n = \chi_n C$ with the character $\chi_n = -\eta_{nn}$, i.e. -1 for the space-like x_0, x_1, x_2, x_3 and $+1$ for the time-like x_4, x_5, x_6, x_7; and $\Gamma P_n = \gamma^n$.

Proof. (1) From $C\gamma^n C^{-1} = -(\gamma^n)^T$ we have $C\gamma^n = -(\gamma^n)^T C$, so

$$(\gamma^n)^T C\gamma^n = -\big((\gamma^n)^T\big)^2 C = -\big((\gamma^n)^2\big)^T C = -\eta_{nn}C.$$

(2) γ^n anticommutes with γ^a for $a \neq n$ and commutes with itself; the second identity follows by inserting the first into $S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$. (3) $\Gamma\gamma^n$ is, up to the sign of the reordering, the product of the eight gammas with γ^n moved to the end, i.e. $\pm\eta_{nn}$ times the product of the other seven; each γ^a is a unit vector ($(\gamma^a)^2 = \eta_{aa} = \pm 1$), so the product lies in Pin(4,4). Using (2), $(\gamma^n)^{-1} = \eta_{nn}\gamma^n$, Lemma 1 and (1),

$$P_n^{-1}\gamma^a P_n = (\gamma^n)^{-1}\Gamma\gamma^a\Gamma\gamma^n = -(\gamma^n)^{-1}\gamma^a\gamma^n = (R_n)^a{}_b\gamma^b,$$
$$P_n^\dagger C P_n = (\gamma^n)^T\Gamma C\Gamma\gamma^n = (\gamma^n)^T C\gamma^n = -\eta_{nn}C,$$

and $\Gamma P_n = \Gamma^2\gamma^n = \gamma^n$. QED. The signs of P_n relative to the product of the other seven gammas (in the order of Γ) are $-1, +1, -1, +1, +1, -1, +1, -1$ for $n = x_0, \ldots, x_7$.

Wolfram records	**sympy records**
`T2_Pn_in_Pin44`, `T2_Pn_covers_the_reflection`, `T2_character_of_Pn`, `T2_Gamma_times_Pn_is_gamma_n`, `T2_kernels_gamma_n_with_frame_reflection`	`T2.general_field.matrix_identities`, `T2.general_field.reflection_table`, `compare.theory.reflection_table`

Lemma 4 (Krein signs of the maps). $MBM^\dagger = \sigma_M B$ with $\sigma_\Gamma = -1$; $\sigma_{\gamma^n} = +1$ for $n = x_0, x_1, x_2, x_3, x_4$ and -1 for $n = x_5, x_6, x_7$; and $\sigma_{P_n} = -\sigma_{\gamma^n}$, i.e. -1 for $P_{x_0}, P_{x_1}, P_{x_2}, P_{x_3}, P_{x_4}$ and $+1$ for $P_{x_5}, P_{x_6}, P_{x_7}$.

Proof. Γ: Lemma 2. For γ^n: it is real, so $(\gamma^n)^\dagger = (\gamma^n)^T = \eta_{nn}\gamma^n$, and transposing $C\gamma^n C^{-1} = -(\gamma^n)^T$ with $C^T = C = C^{-1}$ gives $\gamma^n C = -C(\gamma^n)^T$. Hence

$$\gamma^n B(\gamma^n)^\dagger = -i\gamma^n C\gamma^4(\gamma^n)^T = iC(\gamma^n)^T\gamma^4(\gamma^n)^T = iC\gamma^n\gamma^4\gamma^n.$$

For $n \neq 4$, $\gamma^n\gamma^4\gamma^n = -\eta_{nn}\gamma^4$ and the result is $\eta_{nn}B$; for $n = 4$, $\gamma^4\gamma^4\gamma^4 = -\gamma^4$ and the result is B. Finally $P_n B P_n^\dagger = \Gamma\,\gamma^n B(\gamma^n)^\dagger\,\Gamma = -\sigma_{\gamma^n}B$ by Lemma 2. QED.

Wolfram records	**sympy records**
`Q_Krein_metric_of_images`	`compare.theory.krein_signs`, `Q.image_krein_metric`, `Q.T2_image_keeps_B`

4 Theorem T1: the chirality pairing (both fields)

4.1 Hypotheses

- (H1) The gravitational field: any vielbein $e^a{}_\mu$ (metric $g = e^T\eta e$ of signature (4,4)) and any spin connection $\omega_{\mu ab} = -\omega_{\mu ba}$, in particular the canonical one of the author's metric. The field is a fixed (test) background: it is NOT varied, and both members of the pair are evaluated in the SAME field.
- (H2) The statistics: Ψ Grassmann-valued (dirac16complex) or commuting (dirac16complex00). The map acts at the same point, $\Psi'(x) = \Gamma\Psi(x)$, with no change of coordinates.

- (H3) The potential: $U(S) = \frac{\lambda}{2}S^2$; more generally any function U (commuting field) or any polynomial U (Grassmann field), mapped to $-U$.
- (H4) $\mathcal{L}$, E, T and J as in section 2, with any overall sign or constant factor of T and J.

4.2 Statement

Theorem T1. Under (H1) to (H4), for both fields:

- (T1a) $\mathcal{L}_{m,\lambda}[\Gamma\Psi] = -\mathcal{L}_{-m,-\lambda}[\Psi]$ identically (off shell); for a general potential $\mathcal{L}_{m,U}[\Gamma\Psi] = -\mathcal{L}_{-m,-U}[\Psi]$. Separately, $S[\Gamma\Psi] = S[\Psi]$ and $K[\Gamma\Psi] = -K[\Psi]$.
- (T1b) $E_{m,\lambda}[\Gamma\Psi] = -\Gamma E_{-m,-\lambda}[\Psi]$: $\Gamma\Psi$ solves the field equations with (m,λ) if and only if Ψ solves those with $(-m,-\lambda)$; equivalently, Ψ solves the (m,λ) equations if and only if $\Gamma\Psi$ solves the $(-m,-\lambda)$ equations. The Euler-Lagrange expressions are mapped in the same way.
- (T1c) $T_{\mu\nu}^{(-m,-\lambda)}[\Gamma\Psi] = -T_{\mu\nu}^{(m,\lambda)}[\Psi]$ and $J^\mu[\Gamma\Psi] = -J^\mu[\Psi]$ pointwise, off and on shell.
- (T1d) The pair (Ψ with (m,λ); $\Gamma\Psi$ with $(-m,-\lambda)$) has the total energy-momentum density $T + T' = 0$, the total current $J + J' = 0$ and the total charge $Q + Q' = 0$, as classical bilinears (c-numbers for dirac16complex00, elements of the Grassmann algebra for dirac16complex).

4.3 Proof

Step 1 (bilinears). By Lemma 1, $D_\mu(\Gamma\Psi) = \Gamma D_\mu\Psi$ and $D_\mu(\overline{\Gamma\Psi}) = (D_\mu\bar\Psi)\Gamma$; by Lemma 2, $S[\Gamma\Psi] = S[\Psi]$ and

$$\overline{\Gamma\Psi}\,\gamma^\mu D_\mu(\Gamma\Psi) = \bar\Psi\,\Gamma\gamma^\mu\Gamma\,D_\mu\Psi = -\bar\Psi\gamma^\mu D_\mu\Psi, \qquad (D_\mu\overline{\Gamma\Psi})\gamma^\mu\,\Gamma\Psi = -(D_\mu\bar\Psi)\gamma^\mu\Psi,$$

hence $K[\Gamma\Psi] = -K[\Psi]$. Written out,

$$K = \tfrac{1}{2}\,e^\mu{}_a(\bar\Psi\gamma^a\partial_\mu\Psi - \partial_\mu\bar\Psi\gamma^a\Psi) + \tfrac{1}{4}\,e^\mu{}_a\,\omega_{\mu bc}\,\bar\Psi\{\gamma^a, S^{bc}\}\Psi,$$

a sum of coefficient functions of the gravitational field times bilinears with an odd number of gammas; each such bilinear changes sign by Lemma 2, whatever the coefficients are.

Step 2 (the Lagrangian). $U(S)$ is unchanged because S is, so

$$\mathcal{L}_{m,U}[\Gamma\Psi] = \sqrt{|g|}\,[\,-K - mS - U(S)] = -\sqrt{|g|}\,[K - (-m)S - (-U)(S)] = -\mathcal{L}_{-m,-U}[\Psi],$$

which for $U = \frac{\lambda}{2}S^2$ is (T1a).

Step 3 (the field equations). $\gamma^\mu D_\mu(\Gamma\Psi) = \gamma^\mu\Gamma D_\mu\Psi = -\Gamma\gamma^\mu D_\mu\Psi$, so

$$E_{m,\lambda}[\Gamma\Psi] = -\Gamma\gamma^\mu D_\mu\Psi - (m+\lambda S)\Gamma\Psi = -\Gamma\big[\gamma^\mu D_\mu\Psi - (-m-\lambda S)\Psi\big] = -\Gamma E_{-m,-\lambda}[\Psi],$$

and Γ is invertible, which gives (T1b). For the Euler-Lagrange expressions themselves, in any field and whatever their explicit form: (T1a) is an identity between functionals of Ψ, and $\Psi \mapsto \Gamma\Psi$ is a constant invertible linear substitution, so by the chain rule the Euler-Lagrange expressions of $\mathcal{L}_{m,\lambda}$ at $\Gamma\Psi$, multiplied by Γ, equal minus those of $\mathcal{L}_{-m,-\lambda}$ at Ψ; they vanish together.

Step 4 (energy-momentum tensor and current). $T^{\nu}{}_{\mu}$ is the sum of $\delta^{\nu}_{\mu}\mathcal{L}/\sqrt{|g|}$ and kinetic bilinears with one gamma. By Step 2 with $m \to -m$, $\mathcal{L}_{-m,-\lambda}[\Gamma\Psi] = -\mathcal{L}_{m,\lambda}[\Psi]$, and by Step 1 each kinetic bilinear changes sign; hence $T^{(-m,-\lambda)}[\Gamma\Psi] = -T^{(m,\lambda)}[\Psi]$. The same holds for the tensor of the vielbein variation, whose additional term $\nabla_\lambda(\bar{\Psi}\{\gamma^\lambda, \Sigma^{\nu\rho}\}\Psi)$ contains three gammas. J^μ is a multiple of $\bar{\Psi}\gamma^\mu\Psi$ and changes sign by Lemma 2, and so does $Q = \int \sqrt{|g|}\, J^{x_4}\, d^7x$. This is (T1c).

Step 5 (the pair). Adding (T1c) to the quantities of Ψ gives (T1d).

Step 6 (statistics). $\Gamma = \mathrm{diag}(-I_8, I_8)$ multiplies each component by $+1$ or -1; no Grassmann factors are reordered, so Steps 1 to 5 hold verbatim for anticommuting components, as identities between Grassmann-algebra elements.

The proof uses only Lemmas 1 and 2 and the linearity of $\mathcal{L}$, T and J in the coefficient functions $\sqrt{|g|}$, $e^{\mu}{}_{a}$ and $\omega_{\mu ab}$; hence T1 holds in every gravitational field (H1). QED.

4.4 T1 in the block form of the author's field

In the author's field the proof can be read off the block equations of section 2.5. $\Gamma(\psi_-, \psi_+) = (-\psi_-, \psi_+)$, and C is block diagonal (a product of four gammas; the algebra record gives $C = \mathrm{diag}(-\sigma, \sigma)$ with σ the notebook's 8×8 involution), so $S = -\psi_-^\dagger \sigma \psi_- + \psi_+^\dagger \sigma \psi_+$ is unchanged and $V = m + \lambda S$ becomes $-m - \lambda S = -V$ for the parameters $(-m, -\lambda)$. The two block equations become

$$\sum_a f_a^{-1} \bar{\tau}^a \partial_a \psi_+ + 3H\psi_+ = (-V)(-\psi_-), \qquad -\Big(\sum_a f_a^{-1} \tau^a \partial_a \psi_- + 3H\psi_-\Big) = -V\psi_+,$$

the same equations. None of the gravitational terms ($e^{\mp a_4}\sin^{-1/6} z$, $\tan z$, $3H$) changes.

Wolfram records (author's field)	**sympy records (author's field)**
`T1_Lagrangian_primordial_commuting` (408 monomials), `T1_Lagrangian_primordial_grassmann` (392 monomials), `T1_field_equation_covariance_primordial_commuting`, `T1_field_equation_covariance_primordial_grassmann`, `T1_energy_momentum_primordial_commuting`, `T1_energy_momentum_primordial_grassmann`, `T1_current_primordial_commuting`, `T1_current_primordial_grassmann`	`C_equals_notebook_sigma16`, `T1.metric.commuting.lagrangian`, `T1.metric.grassmann.lagrangian`, `T1.metric.commuting.euler_lagrange_map`, `T1.metric.grassmann.euler_lagrange_map`, `T1.metric.commuting.pair_total_emt_zero`, `T1.metric.grassmann.pair_total_emt_zero`

4.5 Remarks on T1

- Negative controls: $\mathcal{L}_{m,\lambda}[\Gamma\Psi] + \mathcal{L}_{m,\lambda}[\Psi] \neq 0$ and $\mathcal{L}_{m,\lambda}[\Gamma\Psi] + \mathcal{L}_{-m,\lambda}[\Psi] \neq 0$: the mass AND the coupling must change sign. For $\lambda \neq 0$, T1 is therefore not a pure $+m$ / $-m$ pairing; for $\lambda = 0$ (free fields) it pairs $+m$ with $-m$ exactly.
- The partner's energy density, pressures and equations of state: $\rho' = -\rho$ and $p'_\mu = -p_\mu$ in every direction; for homogeneous on-shell states $\rho = mS + \frac{\lambda}{2}S^2$ and $p = \frac{\lambda}{2}S^2$ become $\rho' = -mS - \frac{\lambda}{2}S^2$ and $p' = -\frac{\lambda}{2}S^2$ (with $S' = S$). Every ratio of components, in particular $w_3 = p_3/\rho$, $w_t = p_t/\rho$ and $w_{x_0} = p_0/\rho$, is the same for both members.
- T1 is not a symmetry of one theory: it changes the parameters and the sign of the action. Because $-\mathcal{L}$ and $\mathcal{L}$ have the same Euler-Lagrange equations, $\Gamma\Psi$ is a solution of the $(-m, -\lambda)$ theory; nothing more is asserted.
- Representation content: Γ is -1 on one irreducible Spin(4,4) half and $+1$ on the other (section 2.2); the partner differs from Ψ by the relative sign of its two inequivalent Spin(4,4) components, and Γ is itself an element of Pin(4,4) (a product of eight unit vectors).

Wolfram records (negative controls)	**sympy records (negative controls)**
`T1_Lagrangian_negative_controls_primordial_commuting`, `T1_Lagrangian_negative_controls_primordial_grassmann`, `T1_Lagrangian_negative_controls_diagonal8_commuting`, `T1_Lagrangian_negative_controls_diagonal8_grassmann`, `T1_Lagrangian_negative_controls_pointwise_commuting`, `T1_Lagrangian_negative_controls_pointwise_grassmann`	`T1.metric.commuting.negative_controls`, `T1.metric.grassmann.negative_controls`

5 Theorem T2: the mirror pairing (both fields)

5.1 Hypotheses

- (H5) n is a space-like frame direction (x_0, x_1, x_2 or x_3), and $P_n = \Gamma\gamma^n$ is the Pin(4,4) lift of the reflection R_n, of character -1 (Lemma 3).
- (H6) General-field version: the frame is reflected, $e' = R_n e$ (the same metric, since $R_n \eta R_n = \eta$, and the same $\sqrt{|g|}$), with its canonical connection, and $\Psi'(x) = \Gamma P_n \Psi(x) = \gamma^n \Psi(x)$ at the same point.
- (H7) Author's-field version: the map $\phi : x_0 \mapsto \pi/(6H) - x_0$ ($z \mapsto \pi - z$) takes the patch $z \in (0, \pi/2)$ onto the mirror patch $z \in (\pi/2, \pi)$; each patch carries its own positive vielbein ($e^0 = |\cot z|\, dx_0$, i.e. $-\cot z\, dx_0$ on the mirror patch); and $\Psi'(\phi(x)) = \gamma^0 \Psi(x)$. The brane $z = \pi/2$ is a degenerate surface of the metric ($g_{00} = \cot^2 z = 0$ and $\sqrt{|g|} = b_4^{3/2} \cos z = 0$ there); extending the field to the mirror patch is the ASSUMED Z2 construction of `Revision/SPEC.md` section 7.

- (H8) Both statistics; $U(S) = \frac{\lambda}{2}S^2$ or any even function of S.

5.2 Statement

Theorem T2. Under (H5) to (H8), for both fields:

- (T2a) $\mathcal{L}_{m,\lambda}[\gamma^n\Psi;\, R_n e] = +\mathcal{L}_{-m,\lambda}[\Psi;\, e]$ in every gravitational field; in the author's field $\mathcal{L}_{m,\lambda}[\Psi'](\phi(x)) = \mathcal{L}_{-m,\lambda}[\Psi](x)$.
- (T2b) $S' = -S$, and $E_{m,\lambda}[\gamma^n\Psi;\, R_n e] = -\gamma^n E_{-m,\lambda}[\Psi;\, e]$: Ψ solves the $(-m,\lambda)$ equations if and only if its image solves the (m,λ) equations (in the author's field: on the patch and on the mirror patch respectively).
- (T2c) In the general-field version $T'_{\mu\nu} = T^{(-m,\lambda)}_{\mu\nu}[\Psi]$ and $J'^{\mu} = J^{\mu}[\Psi]$. In the author's field $T_{\mu\nu}[\Psi'](\phi(x)) = (R_0\, T^{(-m,\lambda)}[\Psi](x)\, R_0)_{\mu\nu}$ and $J'^{\mu} = (R_0)^{\mu}{}_{\nu} J^{\nu}$: the energy density and every component without an x_0 index are EQUAL, not opposite. T2 pairs $+m$ with $-m$ at equal energy-momentum.
- (T2d) For a time-like n (character $+1$) the same construction gives $\mathcal{L} \to -\mathcal{L}_{-m,-\lambda}$, a T1-type map; the character -1 is what makes T2 a pairing $(m,\lambda) \to (-m,\lambda)$ with $\mathcal{L} \to +\mathcal{L}$.

5.3 Proof in a general field (frame reflection)

Step 1 (geometry). $g' = e'^T \eta e' = e^T R_n \eta R_n e = g$. The canonical connection of e' is $R_n \omega R_n$: R_n is a constant Lorentz transformation, and the canonical connection $\omega_\mu{}^a{}_b = e^a{}_\nu(\partial_\mu e^\nu{}_b + \Gamma^\nu{}_{\mu\lambda} e^\lambda{}_b)$ is built from the vielbein, its inverse and the Christoffel symbols $\Gamma^\nu{}_{\mu\lambda}$ of the unchanged metric (the symbol with indices is the Christoffel symbol, without indices the chirality matrix). By Lemma 3, $\Omega'_\mu = \frac{1}{2}(R_n \omega_\mu R_n)_{ab} S^{ab} = \gamma^n \Omega_\mu (\gamma^n)^{-1}$, so $D'_\mu(\gamma^n\Psi) = \gamma^n D_\mu \Psi$.

Step 2 (adjoint). By Lemma 3, $(\gamma^n)^T C = -\eta_{nn} C (\gamma^n)^{-1}$, hence $\bar\Psi' = \Psi^\dagger (\gamma^n)^T C = -\eta_{nn} \bar\Psi (\gamma^n)^{-1}$ and $D'_\mu \bar\Psi' = -\eta_{nn}(D_\mu \bar\Psi)(\gamma^n)^{-1}$.

Step 3 (gammas). The reflected frame gives $\gamma'^\mu = e^\mu{}_b (R_n)^b{}_a \gamma^a$, and Lemma 3 gives $\gamma'^\mu \gamma^n = -\gamma^n \gamma^\mu$. Therefore

$$\bar\Psi' \gamma'^\mu D'_\mu \Psi' = -\eta_{nn} \bar\Psi (\gamma^n)^{-1} \gamma'^\mu \gamma^n D_\mu \Psi = \eta_{nn} \bar\Psi \gamma^\mu D_\mu \Psi, \qquad \bar\Psi' \Psi' = -\eta_{nn} \bar\Psi \Psi,$$

and in the same way $(D'_\mu \bar\Psi')\gamma'^\mu \Psi' = \eta_{nn}(D_\mu \bar\Psi)\gamma^\mu \Psi$, so $K' = \eta_{nn} K$ and $S' = -\eta_{nn} S$.

Step 4 (Lagrangian). For space-like n ($\eta_{nn} = +1$): $K' = K$ and $S' = -S$, so

$$\mathcal{L}_{m,\lambda}[\Psi'; e'] = \sqrt{|g|}\,[K + mS - \tfrac{\lambda}{2}S^2] = \mathcal{L}_{-m,\lambda}[\Psi; e],$$

which is (T2a). For time-like n: $K' = -K$ and $S' = S$, so $\mathcal{L}_{m,\lambda}[\Psi'; e'] = -\mathcal{L}_{-m,-\lambda}[\Psi; e]$, which is (T2d).

Step 5 (field equations). For space-like n,

$$E_{m,\lambda}[\Psi';e'] = \gamma'^{\mu} D'_{\mu}(\gamma^n \Psi) - (m + \lambda S')\gamma^n \Psi = -\gamma^n \gamma^{\mu} D_{\mu} \Psi - (m - \lambda S)\gamma^n \Psi = -\gamma^n E_{-m,\lambda}[\Psi; e],$$

which is (T2b).

Step 6 (energy-momentum tensor and current). The kinetic bilinears $\bar{\Psi}\gamma_{\mu} D_{\nu} \Psi$ with free indices transform like the contracted one in Step 3 (factor $\eta_{nn} = +1$), and $\mathcal{L}/\sqrt{|g|}$ transforms by Step 4, so $T'_{\mu\nu} = T^{(-m,\lambda)}_{\mu\nu}[\Psi]$; $\bar{\Psi}'\gamma'^{\mu}\Psi' = \eta_{nn}\bar{\Psi}\gamma^{\mu}\Psi$ gives $J' = J$. This is (T2c) in the general-field version.

Step 7 (statistics). γ^n is a constant signed permutation matrix; the substitution never reorders Grassmann factors, so Steps 1 to 6 hold for both statistics. QED.

5.4 Proof in the author's field (the Z2 mirror)

ϕ is an isometry of the author's metric: its components depend on x_0 only through $\sin z$ and $\cot^2 z$, both invariant under $z \mapsto \pi - z$. The positive vielbein of the mirror patch, pulled back by ϕ, is $R_0 e$: its hidden leg is $-\cot(\pi - z)\, d(\pi/(6H) - x_0) = -\cot z\, dx_0 = -e^0$, and the other legs are unchanged. $\sqrt{|g|}\, d^8x$ is invariant ($b_4^{3/2}|\cos z|$ is invariant and the Jacobian has modulus 1). Pulling the mirror-patch Lagrangian back by ϕ therefore gives the frame-reflected Lagrangian of section 5.3 with $n = x_0$, which proves (T2a) and (T2b) in the author's field; the coordinate components of T and J pick up the Jacobian R_0 of ϕ, which gives (T2c). QED.

In block form γ^0 has the blocks $\bar{\tau}^0 = \tau^0 = I_8$, so $\gamma^0(\psi_-, \psi_+) = (\psi_+, \psi_-)$: the mirror image exchanges the two inequivalent Spin(4,4) halves, and $S = -\psi_-^{\dagger}\sigma\psi_- + \psi_+^{\dagger}\sigma\psi_+$ changes sign.

Wolfram records (author's field)	**sympy records (author's field)**
`T2_mirror_is_isometry`, `T2_mirro r_Lagrangian_commuting`, `T2_mirro r_Lagrangian_grassmann`, `T2_mirro r_energy_momentum_and_current_co mmuting`, `T2_mirror_energy_momentu m_and_current_grassmann`, `T2_mirro r_Euler_Lagrange_commuting`, `T2_mirro r_Euler_Lagrange_grassmann`	`geometry.mirror_isometry`, `geometry .brane_degenerate`, `T2.metric.commut ing.euler_lagrange_map`, `T2.metri c.grassmann.euler_lagrange_map`, `T2.metric.commuting.emt`, `T2.metri c.grassmann.emt`, `T2.metric.commut ing.current`, `T2.metric.grassm ann.current`, `T2.metric.commut ing.S_odd`, `T2.metric.grassmann.S_odd`

5.5 The eight reflections

The sympy checker also shows that neither the plain mirror ($P = 1$) nor Γ alone gives a relation $\mathcal{L}_{\text{mirror}} = \pm\mathcal{L}_{\text{patch}}$ with $(m, \lambda) \to (\pm m, \pm\lambda)$, that $P_0 = \Gamma\gamma^0$ alone gives $\mathcal{L} \to -\mathcal{L}_{m,-\lambda}$, and that its composition with Γ gives T2. The frame reflections of all eight directions, from the data of the pairing theory record (the Wolfram pointwise frame-reflection checks of every direction and the sympy frame-reflection checks of the general diagonal field, both statistics; the record names are in the T2 tables of section 9.2):

direction	η_{nn}	**character of** P_n	S **under** γ^n	**kinetic term**	**map**
x_0	$+1$	-1	$-S$	$+K$	$(-m,\lambda)$, $+\mathcal{L}$ (T2)
x_1	$+1$	-1	$-S$	$+K$	$(-m,\lambda)$, $+\mathcal{L}$ (T2)
x_2	$+1$	-1	$-S$	$+K$	$(-m,\lambda)$, $+\mathcal{L}$ (T2)
x_3	$+1$	-1	$-S$	$+K$	$(-m,\lambda)$, $+\mathcal{L}$ (T2)
x_4	-1	$+1$	$+S$	$-K$	$(-m,-\lambda)$, $-\mathcal{L}$ (T1-type)
x_5	-1	$+1$	$+S$	$-K$	$(-m,-\lambda)$, $-\mathcal{L}$ (T1-type)
x_6	-1	$+1$	$+S$	$-K$	$(-m,-\lambda)$, $-\mathcal{L}$ (T1-type)
x_7	-1	$+1$	$+S$	$-K$	$(-m,-\lambda)$, $-\mathcal{L}$ (T1-type)

In the author's field only the reflection of x_0 is realised by an isometry that maps the patch onto another patch (the mirror across the Z2 brane); the reflections of x_1, x_2, x_3 are frame statements.

6 The quantum-level reading (dirac16complex)

6.1 Hypotheses

- (HQ1) dirac16complex is canonically quantised with x_4 as the evolution time. The x_4-derivative kernel of $\mathcal{L}_{m,\lambda}$ is $\frac{i}{2}\sqrt{|g|}\,N$ with $N = B$ (since $C\gamma^4 = iB$), and the canonical anticommutator on a slice $x_4 = \text{const}$ is $\{\Psi_A(x), \Psi^\dagger_B(y)\} = (N^{-1})_{AB}\,\delta^7(x-y)/\sqrt{|g|} = B_{AB}\,\delta^7(x-y)/\sqrt{|g|}$.
- (HQ2) dirac16complex00 is a classical field and has no quantum reading.

6.2 Statements

- (Q1) The chirality image $\chi = \Gamma\Psi$ carries the Krein metric $-B$: $\{\chi_A, \chi^\dagger_B\} = (\Gamma B \Gamma)_{AB}\,\delta/\sqrt{|g|} = -B_{AB}\,\delta/\sqrt{|g|}$, which is also the anticommutator demanded by its own Lagrangian $\mathcal{L}_{m,\lambda}[\Gamma\chi] = -\mathcal{L}_{-m,-\lambda}[\chi]$.
- (Q2) The image's own generators (energy, momentum and charge from its own Lagrangian) coincide with those of Ψ: the pair Ψ, $\Gamma\Psi$ is ONE quantum system, and the T1 identity $T^{(-m,-\lambda)}[\Gamma\Psi] = -T^{(m,\lambda)}[\Psi]$ is an identity between operators of that one system.
- (Q3) An independently quantised $(-m,-\lambda)$ universe has the anticommutator $+B$ (from $\mathcal{L}_{-m,-\lambda}$), its own vacuum and its own generators; it cannot be identified with $\Gamma\Psi$ ($B \neq -B$); on the product state space the generators add, $P_{\text{total}} = P_1 \otimes 1 + 1 \otimes P_2$, and no cancellation $P_1 + P_2 = 0$ follows.

- (Q4) The one-particle ($\lambda = 0$) Hamiltonians satisfy $\Gamma h_m \Gamma = h_{-m}$ (flat space, and a general field at a point with frozen coefficients) and $\gamma^0 h_m(k)\gamma^0 = h_{-m}(R_0 k)$: the $+m$ and $-m$ one-particle spectra are IDENTICAL, not opposite. In flat 4+4 space $h_m(k) = -im\gamma^4 - \gamma^4 \sum_{a\neq 4} k_a \gamma^a$ and $h_m(k)^2 = w^2 I_{16}$ with $w^2 = m^2 + k_0^2 + k_1^2 + k_2^2 + k_3^2 - k_5^2 - k_6^2 - k_7^2$; for every REAL frequency ($w^2 > 0$, with or without extra-time momentum) each of the two eigenspaces has dimension 8 and Krein inertia (4,4), the same for $+m$ and $-m$, and for imaginary or zero frequency (the growing extra-time modes) the eigenspaces are Krein-neutral (B vanishes on them).
- (Q5) The T2 image $\gamma^0\Psi$ keeps the anticommutator $+B$: the mirror $(-m, \lambda)$ universe is an ordinary, independently quantisable copy with equal energies.

6.3 Proofs

(Q1) By Lemma 2, $\{\Gamma\Psi, (\Gamma\Psi)^\dagger\} = \Gamma\{\Psi, \Psi^\dagger\}\Gamma^\dagger = \Gamma B \Gamma\, \delta/\sqrt{|g|} = -B\,\delta/\sqrt{|g|}$. For the image's own Lagrangian, the x_4-velocity term $\frac{i}{2}\sqrt{|g|}\,\chi^\dagger \Gamma B \Gamma\, \partial_4 \chi$ has the kernel $N = -B$, hence the anticommutator $N^{-1} = -B$: the two agree.

(Q2) At $\chi = \Gamma\Psi$ the image Lagrangian $\mathcal{L}_{m,\lambda}[\Gamma\chi]$ is $\mathcal{L}_{m,\lambda}[\Psi]$, the same function of the same operators, so its Legendre energy density and its Noether densities coincide with those of Ψ; they are minus the generators of the $(-m, -\lambda)$ Lagrangian evaluated at χ. With the metric $-B$ and the generator density $-h_m$ the image evolves by $(-B)(-h_m) = B h_m$, exactly the (m, λ) dynamics.

(Q3) The Lagrangian $\mathcal{L}_{-m,-\lambda}[\chi_2]$ of an independent universe has the kernel $N = +B$, hence $\{\chi_2, \chi_2^\dagger\} = +B\,\delta/\sqrt{|g|}$, while $\{\Gamma\Psi_1, (\Gamma\Psi_1)^\dagger\} = -B\,\delta/\sqrt{|g|}$; and independence requires $\{\Psi_1, \chi_2^\dagger\} = 0$, whereas $\{\Gamma\Psi_1, \Psi_1^\dagger\} = \Gamma B/\sqrt{|g|}$ has rank 16. So no identification $\chi_2 = \Gamma\Psi_1$ exists. On the product space the generators of the two systems add; the T1 identity relates operators of universe 1 only and gives no relation $P_2 = -P_1$. At zero momentum the block one-particle generator $\mathrm{diag}(Bh_m, Bh_{-m})$ has the eigenvalues $-m$ and m, each 16 times, all nonzero for $m \neq 0$.

(Q4) $\Gamma\gamma^4\Gamma = -\gamma^4$ and $\Gamma\gamma^4\gamma^a\Gamma = \gamma^4\gamma^a$ give $\Gamma h_m \Gamma = h_{-m}$. With $(\gamma^0)^2 = 1$,

$$\gamma^0\gamma^4\gamma^0 = -\gamma^4, \qquad \gamma^0\gamma^4\gamma^a\gamma^0 = \gamma^4\gamma^a \ (a \neq 0, 4), \qquad \gamma^0\gamma^4\gamma^0\gamma^0 = -\gamma^4\gamma^0,$$

which gives $\gamma^0 h_m(k)\gamma^0 = h_{-m}(R_0 k)$. Similar matrices have equal spectra. The square and the eigenspaces are computed exactly; the samples of `Revision/pairing/pairing-theory.json` are:

m	**momentum** $(k_0, \ldots, k_7)$	w	**dimensions of the $+w$ and $-w$ eigenspaces**	**Krein inertia on $+w$ and on $-w$**
2	(4, 1, 2, 0, 0, 0, 0, 0)	5	8 and 8	(4,4) and (4,4)
−2	(4, 1, 2, 0, 0, 0, 0, 0)	5	8 and 8	(4,4) and (4,4)
3	(0, 0, 0, 0, 0, 0, 0, 0)	3	8 and 8	(4,4) and (4,4)
−3	(0, 0, 0, 0, 0, 0, 0, 0)	3	8 and 8	(4,4) and (4,4)
1	(0, 1, 0, 0, 0, 0, 0, 0)	$\sqrt{2}$	8 and 8	(4,4) and (4,4)

m	**momentum** $(k_0, \ldots, k_7)$	w	**dimensions of the $+w$ and $-w$ eigenspaces**	**Krein inertia on $+w$ and on $-w$**
-1	(0, 1, 0, 0, 0, 0, 0, 0)	$\sqrt{2}$	8 and 8	(4,4) and (4,4)
2	(0, 0, 0, 0, 0, 1, 0, 0)	$\sqrt{3}$	8 and 8	(4,4) and (4,4)
-2	(0, 0, 0, 0, 0, 1, 0, 0)	$\sqrt{3}$	8 and 8	(4,4) and (4,4)

The entry k_4 is not used (h_m contains no $\gamma^4\gamma^4$ term). The last two rows have an extra-time momentum and a real frequency. The general statement is proved, not sampled: (i) for real $w > 0$ the projectors $P_\pm = (1 \pm h_m/w)/2$ satisfy $P_+^\dagger B P_- = 0$ and $P_+^\dagger B P_+ = B P_+$ (from $B h_m = h_m^\dagger B$ and $h_m^2 = w^2$), so the two eigenspaces are B-orthogonal and B is nondegenerate on each; (ii) in the good sector $[B, h_m] = 0$ and $\operatorname{tr} B = \operatorname{tr}(B h_m) = 0$, so $\operatorname{tr}(B P_\pm) = 0$ and B has inertia (4,4) on each eigenspace; (iii) the region $w^2 > 0$ is connected and contains the good sector (lowering k_5, k_6, k_7 to zero only increases w^2), and $P_\pm$ depend continuously on k there, so the inertia (4,4) holds at every real frequency. For imaginary w, $hu = wu$ and $hv = wv$ give $(w - \bar{w})\, u^\dagger B v = 0$: the eigenspace is Krein-neutral; for $w = 0$, $h^2 = 0$ with rank 8 and $\ker h = \operatorname{ran} h$ is B-neutral. Exact samples: $m = 1$, $k_5 = 2$ ($w = \pm i\sqrt{3}$); $m = 1$, $k_1 = 1$, $k_5 = 2$ ($w = \pm i\sqrt{2}$); $m = 1$, $k_5 = 1$ ($w = 0$): every eigenspace has dimension 8 and the form $u^\dagger B v$ vanishes on it identically. In a general field at a point the same similarity $\Gamma h_m \Gamma = h_{-m}$ holds for symbolic momenta; in the pointwise test field $(\gamma^{x_4})^2 = g^{x_4 x_4} I_{16} = -\frac{2307429}{1990921} I_{16}$ (the coefficients of that field are a formula in the list positions of the coordinates; since the relabel of 2026-10-14 the positions are those of the author's order, so it is a different, equally generic field: $\det e$ and the number of nonzero connection components are unchanged, $g^{x_4 x_4}$ was $-\frac{2269482}{1990921}$ before).

(Q5) Lemma 4 gives $\gamma^0 B (\gamma^0)^\dagger = +B$. QED.

statement	**Wolfram records**	**sympy records**
(HQ1)	`first_order_form_and_anticommutator`	`canonical_anticommutator_B`, `Q.canonical_anticommutator`
(Q1)	`Q_Krein_metric_of_images`, `Q_symplectic_kernel_commuting`, `Q_symplectic_kernel_grassmann`	`Q.image_krein_metric`, `Q.image_own_quantisation`
(Q2)	`Q_generators_of_the_image_commuting`, `Q_generators_of_the_image_grassmann`	`Q.image_generators_same_dynamics`
(Q3)	`Q_no_identification_of_independent_universes`	`Q.no_cancellation_independent_universes`

statement	Wolfram records	sympy records
(Q4)	Q_one_particle_flat_dispersion, Q_one_particle_maps, Q_one_particle_Krein_signatures, Q_one_particle_Krein_inertia_real_frequencies, Q_one_particle_complex_and_zero_frequencies_Krein_neutral, Q_one_particle_general_field	Q.one_particle_maps, Q.one_particle_Krein_inertia, Q.one_particle_Krein_inertia_proof, Q.one_particle_complex_frequency_Krein_neutral, compare.theory.one_particle
(Q5)	Q_Krein_metric_of_images	Q.T2_image_keeps_B

Relation to the good sector of the theory record. With $\Psi^\dagger$ realised as the Hilbert adjoint, the canonical anticommutator forces an indefinite (Krein) inner product (record `no_positive_inner_product`). The theory record constructs, in the good sector without extra-time momentum, a positive Fock representation in which $\Psi^\dagger$ is realised as χB with χ the Hilbert adjoint, but only for single good-sector momenta with frozen coefficients (flat-frame plane waves, in which the deflation of the extra times does not enter; records `mode_hamiltonian_good_sector`, `Fock_space_good_sector_example` and `good_sector_positive_fock_realisation`). In the author's metric the curved good-sector mode operator is Hermitian only up to a boundary term at $z = \pi/2$ (`good_sector_hermiticity_up_to_the_brane_flux`); the theory record imposes no boundary condition there, and without one the good sector has complex frequencies, for $U = 0$ whenever $m^2 < 9H^2$ (`good_sector_x0_independent_modes_without_boundary_condition`). A positive-norm Hilbert space for the full field is not established. The statements Q1 to Q5 are statements about the canonical anticommutator; they neither use nor establish a positive-norm Fock space for either universe.

7 Corollary C1: a T1 pair as the source of the field equations for a_4 and b_4

Hypotheses. (i) The gravitational field is the author's metric with $b_4(x_4)$ and the Einstein-Lovelock equations $\sum_{k=1}^{3} \alpha_k E_{(k)}{}^{\mu}{}_{\nu} + \Lambda \delta^{\mu}_{\nu} = \kappa T^{\mu}{}_{\nu}$ of `Revision/SPEC.md` section 5. (ii) The complete source is the sum of the classical energy-momentum tensors of the two members of a T1 pair at the same points of the same patch (dirac16complex00, or dirac16complex as a Grassmann-algebra-valued classical bilinear). (iii) Nothing else sources the metric.

Statement. The total source vanishes identically, so a_4 and b_4 must solve the vacuum equations. (a) In Einstein gravity ($\alpha_1 = 1$, $\alpha_2 = \alpha_3 = 0$) there is no real solution for $H > 0$ and any Λ: the x_4-x_0 component forces $b_4' = 0$, and then the x_4 and x_0 components give $36H^2 + 2\Lambda = 0$ and $6(a_4')^2 + 6H^2 = 0$. (b) With the author's Lovelock weights ($\alpha_k = w_k/H^{2k}$, $w = (1, \frac{1}{32}, \frac{1}{1152})$, $\Lambda = -10H^2$) an exponential history $a_4 = AHx_4 + a_0$, $b_4 = b_4(0)e^{\beta AHx_4}$ is a vacuum solution exactly for $\beta = 0$ (b_4 constant, the withdrawn metric) and $A = \pm 1$; none with b_4 not constant. (c) In Einstein-Gauss-Bonnet gravity ($\alpha_1 = 1$, $\alpha_3 = 0$) the vacuum exponential histories with

b_4 not constant are exactly $\beta = 4$ with $\alpha_2 = 1/(40H^2(1-A^2))$ and $\beta = 2(1+A^2)/A^2$ with $\alpha_2 = -A^2/(4H^2(A^2-1)^2) < 0$ ($A^2 \neq 1$): tuned couplings.

Proof. (T1d) gives $T + T' = 0$, so the right-hand side of the field equations is zero, and (a), (b) and (c) are the vacuum statements of `Revision/field_equations_a4` (records below). QED.

statement	Wolfram records	sympy records
(a)	`einstein_no_vacuum_solution`, `einstein_offdiagonal_forces_b4_constant`	`einstein_no_vacuum`
(b)	`author_weights_no_vacuum_with_b4_not_constant`	`author_weights_no_vacuum_with_b4_not_constant`
(c)	`egb_exponential_vacuum_branches`	`egb_exponential_vacuum_branches`

Consequence (exact, under (i) to (iii)): a T1 pair cannot by itself be the source of the author's metric in Einstein gravity, nor, with the author's weights, of any exponential history with b_4 not constant. This is a statement about sources in one common geometry; it is not a derivation that such a geometry, or such a pair, is created. For a T2 pair the sources do not cancel: the mirror partner carries the pulled-back energy-momentum tensor $R_0 T R_0$ of the $(-m, \lambda)$ configuration. For two independently quantised universes C1 does not apply (Q3).

8 Theorem T3: the Kohn-Sham level (dirac16complex)

`Revision/SPEC.md` section 9 asks for T3: at the Kohn-Sham level, the block map with the transformed boundary conditions maps the instantaneous Kohn-Sham problem with (m, λ) to the one with $(-m, +\lambda)$ with equal energies. It is proved by two independent exact verifiers in `Revision/pairing/kohn_sham/` (theorem record `Revision/pairing/kohn_sham/t3-theory.json`).

8.1 Hypotheses

- (H3.1) The problem: the instantaneous Kohn-Sham problem of `Revision/kohn_sham/ks-theory.json` at a fixed slice $a_{4,0} = a_4(x_4)$, $b_{4,0} = b_4(x_4)$ of the author's metric with $b_4(x_4)$ (b_4 does not enter the block Hamiltonians; it enters the proper densities only through the cell volume): good sector (no extra-time momentum), a lattice of 3-space momenta k, and in each of the eight 2×2 blocks the Hamiltonian $h_j = j[-i\sigma_1 \, d/dy + M_{\text{eff}}(y)\sigma_2 + \kappa(y)k\sigma_3] + v_v(y)$ on $y \in [-L, 0]$ ($j = \pm 1$; four blocks (s_2, s_3) per j).
- (H3.2) The functional: Hartree plus the exact uniform-gas exchange, $M_{\text{eff}} = m + \frac{15}{16}\lambda S$, $v_v = -\lambda n/16$, $e_{\text{int}} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2$, no correlation; the energy-momentum tensor of that record.
- (H3.3) The brane (ASSUMED): the Z2 mirror (orbifold) at $y = 0$; both brane parities, $\chi_2(0) = 0$ and $\chi_1(0) = 0$, are solved and filled together.
- (H3.4) The tip (chosen): $(1 - Q(\theta))\chi(-L) = 0$ with $Q(\theta) = \cos\theta \, \sigma_3 + \sin\theta \, \sigma_2$.

- (H3.5) Mermin occupations at fixed particle number N and temperature T, with the filling convention of the Kohn-Sham record (positive branch of the $\lambda = 0$ problem plus the $k = 0$ zero modes, followed continuously in λ).
- (H3.6) The same H, L, $a_{4,0}$, $b_{4,0}$, lattice, extra-time volume, N and T for both members; quasi-free (mean-field) states.

8.2 Statement

For every self-consistent Kohn-Sham state with (m, λ, θ), the map $(\chi, j) \to (\sigma_2\chi, -j)$ at the same k, which is the chirality Γ of the 16-component orbital in the block basis (up to the phase s_2 per block), with the two brane parities exchanged, gives a self-consistent Kohn-Sham state with $(-m, +\lambda, \pi - \theta)$, and conversely (the map is an involution). The two states have the same levels with their degeneracies, the same occupations, chemical potential, particle number and entropy, EQUAL Kohn-Sham energy E_{KS}, grand potential and free energy, and equal energy-momentum profiles $\rho(y)$, $p_3(y)$, $p_t(y)$, $p_0(y)$; the densities transform as $n \to n$, $S \to -S$, $Q \to -Q$, so $M_{\text{eff}} \to -M_{\text{eff}}$ and $v_v \to v_v$. With the untransformed tip angle the spectra differ. Inside the ASSUMED Z2 orbifold the mirror copy of a self-consistent state carries $(-m, +\lambda)$.

8.3 Proof

1. $\sigma_2 h_j(M, k, v)\sigma_2 = h_{-j}(-M, k, v)$ as differential operators, and for the shooting form $\chi' = N\chi$, $N = M\sigma_3 - \kappa k\sigma_2 + ij(\varepsilon - v)\sigma_1$: $\sigma_2 N_j(M)\sigma_2 = N_{-j}(-M)$. So $\sigma_2\chi$ is an eigen-orbital of block $-j$ with $-M$ at the same level ε.
2. $\sigma_2 Q(\theta)\sigma_2 = Q(\pi - \theta)$ and $\sigma_2(1 - \sigma_3)\sigma_2 = 1 + \sigma_3$: the tip condition goes into the one with $\pi - \theta$ (the canonical $\theta = 0$ into $\theta = \pi$) and the brane parities are exchanged; the boundary current $-ij\,\phi^\dagger\sigma_1\chi$ and the norm are invariant, so the image problem is self-adjoint with the same normalisation.
3. Per orbital, n_o and t_o are invariant and s_o, q_o change sign; hence $n(y)$ is unchanged and $S(y)$ changes sign. $M_{\text{eff}}[-m, +\lambda, -S] = -M_{\text{eff}}[m, \lambda, S]$, v_v and e_{int} are even in S: the self-consistency loop commutes with the map exactly for $(-m, +\lambda)$ (not for $(-m, -\lambda)$).
4. The map is a bijection between the eigen-orbitals of the two problems with equal levels (both j and both parities belong to each problem), so occupations, μ, N and entropy agree; every term of E_{KS}, Ω, F and of the energy-momentum profiles is a sum of invariant densities, or contains the product $M_{\text{eff}} s_o$ of two odd factors, plus the even e_{int}. QED.

Independent confirmations inside the same records: for $k = 0$, $v = 0$ and constant M the characteristic functions of $(M, j, \theta = 0)$ and $(-M, -j, \theta = \pi)$ agree parity by parity, so the two spectra are equal level by level, while the untransformed tip gives a different characteristic function; the 2×2 map is the 16-component Γ in the block basis V of the Kohn-Sham record; the Z2 mirror $P_A\chi(y) = \sigma_3\chi(-y)$ makes the doubled problem symmetric exactly when the mirror copy carries $(-m, +\lambda)$.

step	Wolfram records	sympy records
1	T3_block_hamilto nian_map, T3_ode_map	T3.block_hamilto nian_map, T3.ode_map
2	T3_tip_condition_map, T3_brane_paritie s_exchanged	T3.tip_condition_map, T3.brane_paritie s_exchanged
3	T3_orbital_densities, T3_mean_field_map	T3.orbital_densities, T3.mean_field_map
4	T3_energies_and_ emt_profiles_equal	T3.energies_and_ emt_profiles_equal
confirmations	T3_exact_k0_spectra, T3_Gamma_is_the_ block_map, T3_z2_mi rror_copy_carrie s_minus_m_plus_lambda	T3.exact_k0_spectra, T3.Gamma_is_the_ block_map, T3.z2_mi rror_copy_carrie s_minus_m_plus_lambda
comparison with the theorem record, numerical confirmation		compare.t3_theor y.theorem, compare. t3_theory.not_es tablished, T3.rust_ selftest_numeric al_confirmation

The ingredients recorded earlier by the Kohn-Sham theory reports agree with these steps: `block_Ga mma_map`, `blocks_relation_to_Gamma`, `bc_mirror_map_PA`, `bc_mirror_parities_of_densities` and `bc_brane_parity_conditions` (both Kohn-Sham theory reports).

8.4 Numerical confirmation and interpretation

A numerical self-test of the Rust solver, labelled in its report as NOT a proof of T3: $(m, \lambda, \text{tip } \theta = 0)$ and $(-m, \lambda, \text{tip } \theta = \pi)$, solved independently, give the same sorted levels, occupations, Kohn-Sham energies and energy-momentum integrals and opposite S (worst deviation 8.35e-14, tolerance 1e-9). For $N = 8$, $\lambda = 0.01946$ and $a_{4,0} = 1$: E_{KS} = -1.341747939926e-4 versus -1.341747939933e-4 (80 levels), while the negative control with the untransformed tip $b(-L) = 0$ gives E_{KS} = -1.6232217540e0. For $N = 136$ and $\lambda = 0.01206$: E_{KS} = 3.239983783344e1 in both runs (160 levels), negative control 2.9056626386e1. The couplings are the solvers' $\hat{\lambda}$ and the units are those of `Revision/kohn_sham/ks-theory.json`.

Interpretation (labelled, not a theorem): in its parameters the Kohn-Sham-level map is of the T2 type, $(m, \lambda) \to (-m, +\lambda)$ at equal energies, not of the T1 type, although its 16-component matrix is the chirality Γ of T1: the Kohn-Sham expectation values are taken with the fixed Krein metric B of the canonical quantisation, under which Γ flips the sign ($\Gamma B \Gamma = -B$, statement Q1), so $S \to -S$ at the Kohn-Sham level while $S[\Gamma\Psi] = S[\Psi]$ for the classical bilinear.

What T3 does not establish: no creation process, rate or amplitude; only instantaneous (adiabatic) mean-field states (the time-dependent problem is OPEN); the Z2 brane is ASSUMED and the tip angle must be transformed; no correlation; no statement about two independently quantised universes; no back-reaction (the Kohn-Sham energy-momentum tensor does not satisfy the source conditions of the a_4 and b_4 equations, records `ks_profiles_depend_on_x0` and `ks_profiles_viol ate_algebraic_condition`). The a_4 and b_4 equations with the Kohn-Sham source (`Revision`

/field_equations_a4/ks_source/reports/ks-source-a4.json, 22 of 22 checks) confirm this exactly: the recorded states conserve energy-momentum along x_0 (B2_x0_conservation_on_every_profile), so their diagonal components depend on x_0, which the field equations exclude for every a_4 and b_4, and the $x_1 + x_5 - 2x_0$ condition of the history fails also after averaging over the hidden direction (C1_averaged_algebraic_condition_fails); the coupled problem has no solution, so nothing about a_4 or b_4 is derived from it, and the integration of a truncated, averaged system there is an approximation with residuals of order one, not used here.

8.5 Completion of T3 (2026-10-08)

An adversarial verification of T3 (2026-10-08) re-ran both T3 verifiers, which reproduce the theorem record t3-theory.json and the two T3 reports byte for byte, and re-derived every step of the proof; no error was found in the statement or the proof. Three gaps were found. A completion closes them by exact computation in a separate record, t3-completion.json, written by the Wolfram verifier verify_t3_completion.wls and re-derived independently by the sympy checker check_t3_completion.py (all in the folder Revision/pairing/kohn_sham; the reports lie in its subfolder reports/); T3 and its statements S1 to S5 are unchanged.

1. At the adversarial verification the Kohn-Sham coefficients 15/16, −1/16, 15/32, −1/32 of (H3.2) were written into the T3 verifier instead of being read from the Kohn-Sham record, and one of its clauses about v_v compared an expression with itself. The completion reads the potentials from Revision/kohn_sham/ks-theory.json, checks $M_{\text{eff}} - m = \partial e_{\text{int}}/\partial S$ and $v_v = \partial e_{\text{int}}/\partial n$, and repeats the map of step 3 with them ($M_{\text{eff}}[-m, +\lambda, -S] = -M_{\text{eff}}[m, \lambda, S]$; with $(-m, -\lambda)$ it fails, the control). The T3 verifier itself was then fixed at the root (2026-10-08): it reads the coefficients from the Kohn-Sham record, and its check T3_mean_field_map compares v_v as a function of (m, λ, n, S) at the original and the image arguments (still 10 of 10).

2. The filling convention (H3.5) was used for the $-m$ member without showing that its particle set is the image of the $+m$ member's. The completion shows that the map sends the $\lambda = 0$ levels with $\varepsilon > 0$ onto those of the image problem, commutes with the continuation in λ, and takes the $k = 0$ brane zero mode to the brane zero mode of the image in its own boundary conditions (odd parity, tip $\theta = \pi$); with the untransformed tip the image mode violates the tip condition (control). So the occupied set of the image is the image of the occupied set.

3. The proof covers the four diagonal energy-momentum components of the 2×2 reduction. Statement S6 extends it to the 16-component level: with the expectation rule $\rho = \sum f\, uu^\dagger B$ of the canonical quantisation and $\Gamma B \Gamma = -B$, the image state has $\rho' = -\Gamma\rho\Gamma$, so n is unchanged, S and Q change sign, and every component of the 16-component energy-momentum tensor and of the current J^a of the $(-m, +\lambda)$ image state equals that of the original state.

gap	Wolfram records	sympy records
1	T3C_mean_field_coefficients_from_ks_theory	T3C.mean_field_coefficients_from_ks_theory
2	T3C_filling_convention_mapped	T3C.filling_convention_mapped

gap	Wolfram records	sympy records
3 (S6) T3 reports, the two demonstrations, comparison with the completion record	`T3C_krein_rule_1 6_component`	`T3C.krein_rule_1 6_component` `T3C.t3_reports_pass`, `T3C.rust_demo_numerical_ demonstration`, `T3C.refe rence_demo_numer ical_demonstration`, `compare.t3_completion`

The numerical demonstrations (not part of the proof) solve, at the slices $a_{4,0} = 0, 0.5, 1, 1.5, 2$ of the prescribed background history $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$ ($A = 1$, the extra times deflating as $e^{-a_4}b_4^{1/2} = b_4(0)^{1/2}e^{-a_4/3}$; the coupling $\hat{\lambda}$ of each state is held fixed while the cell volume grows), three members of each state: the $+m$ universe $(m, \lambda, \theta = 0)$, the $-m$ universe $(-m, +\lambda, \theta = \pi)$ and a control.

- Rust solver (report `t3-rust-demo.json`, 210 states: all 75 ground and 135 Mermin states of the canonical matrix of the Kohn-Sham record): the $+m$ members reproduce the committed canonical matrix to 2.538e-10; the image equals the $+m$ state to 1.741e-13 (ground) and 2.120e-11 (thermal), tolerance 1e-9, label by label with the sector map $(n_2, j, \text{parity}) \to (n_2, -j, \text{other parity})$, including the energy-momentum integrals and profiles, with S, Q, M_{eff} of opposite sign (`t3_equal_ground_states`, `t3_equal_thermal_states`). The shooting method is covariant under the map, so this equality is at the rounding level by construction; it tests the solver's handling of the transformed boundary conditions, the labels, the filling convention, the self-consistency and the Mermin root for both members. Controls: with the untransformed tip the density profile differs from the $+m$ state's by at least 5.296e-01 of its maximum (`negative_control_untransformed_tip`; 200 states with a converged control; in 10 states, all with $\lambda < 0$, its self-consistent field does not converge, which is reported as such and is not a proof that no such state exists); the T1 parameters $(-m, -\lambda, \theta = \pi)$ differ from the $+m$ state by at least 3.254e-04 in all 150 states with $\lambda \neq 0$ (`negative_control_lambda_ sign`): the Kohn-Sham partner carries $+\lambda$.
- Independent reference solver (report `t3-reference-demo.json`, 18 states, finite differences with Richardson extrapolation): the image equals the $+m$ state to 9.326e-15 in the ground states after extrapolation (tolerance 1e-9; the Mermin states pass the same test), while on a single grid the two discretisations differ by up to 2.24e-04, so this equality is a continuum statement; with the untransformed tip the density profile differs by at least 5.296e+00 of its maximum; the $-m$ universes of the two solvers agree to 1.262e-11 (`reference_image_ equals_rust_image`).

What the completion does not establish: the same limits as T3 (section 8.4); in addition the filling convention remains a CONVENTION whose justification is open, and S6 is an equality of expectation values in Kohn-Sham (quasi-free) states with the fixed Krein metric B, not a statement about two independently quantised universes (the list of what is not established in the completion record).

9 Verification records

9.1 Reports and their counts

Every check has a name, a verdict and a detail. Counts at the time of writing (the publication test re-reads them):

report	checks	PASS	FAIL
Revision/pairing/reports/wolfram-pairing.json	102	102	0
Revision/pairing/reports/python-pairing.json	67	67	0
Revision/pairing/kohn_sham/reports/wolfram-t3.json	10	10	0
Revision/pairing/kohn_sham/reports/python-t3.json	13	13	0
Revision/pairing/kohn_sham/reports/wolfram-t3-completion.json	3	3	0
Revision/pairing/kohn_sham/reports/t3-rust-demo.json	7	7	0
Revision/pairing/kohn_sham/reports/t3-reference-demo.json	7	7	0
Revision/pairing/kohn_sham/reports/python-t3-completion.json	7	7	0
Revision/algebra/reports/wolfram-algebra.json	45	45	0
Revision/algebra/reports/python-algebra.json	35	35	0
Revision/theory/reports/wolfram-field-theory.json	84	84	0
Revision/theory/reports/python-field-theory.json	72	72	0
Revision/theory/reports/wolfram-scope.json	15	15	0

report	checks	PASS	FAIL
Revision/theory/reports/python-scope.json	14	14	0
Revision/field_equations_a4/reports/wolfram-a4-report.json	59	59	0
Revision/field_equations_a4/reports/python-a4-report.json	65	65	0
Revision/field_equations_a4/ks_source/reports/ks-source-a4.json	22	22	0
Revision/kohn_sham/reports/ks-theory-wolfram.json	51	51	0
Revision/kohn_sham/reports/ks-theory-python.json	63	63	0
Revision/kohn_sham/reports/ks-rust-solver.json	43	43	0

The pairing theorems are verified in four gravitational fields: the author's metric on the patch $z \in (0, \pi/2)$ (primordial); the same metric on the mirror patch; a general diagonal (4,4) field $e^a{}_\mu = h_a(x_0, \ldots, x_7)\delta^a_\mu$ with eight arbitrary functions of all eight coordinates (diagonal8); and a general non-diagonal field at one point (pointwise), with exact rational $e^a{}_\mu$ and $\partial_\rho e^a{}_\mu$, $\det e = 1411/6561$, a non-diagonal metric and 448 of the 512 connection components nonzero; plus the linearity (kernel) argument that covers every field. The canonical connection of each test field satisfies the vielbein postulate exactly. The Wolfram side and the sympy side share no code; the sympy side re-builds the gammas from the author's tau formulas and compares them entry by entry with `Revision/algebra/gammas.json`.

9.2 Checks per theorem

The tables below list all 102 checks of the Wolfram pairing report and all 67 checks of the sympy pairing report (section 9.1), grouped by theorem; every one has the verdict PASS (the checks of T3 are listed in section 8.3). Among the common sympy checks, the three comparison checks for T1, T2 and Q record that the sympy side confirms each Wolfram theorem independently (with 17, 13 and 10 named sympy checks), and the comparison check for the limits records that the two independently written lists of what the theorems do not establish cover the same topics; section 11.2 merges them.

T1, Wolfram (51 checks):

check	check	check
T1_kernel_scalar	T1_kernel_kinetic	T1_kernel_connection
T1_kernel_field_ equation	T1_linearity_argument	T1_scalar_and_ki netic_primordial _commuting
T1_Lagrangian_pr imordial_commuting	T1_Lagrangian_ne gative_controls_ primordial_commuting	T1_field_equatio n_covariance_pri mordial_commuting
T1_Euler_Lagrang e_derived_primor dial_commuting	T1_Euler_Lagrang e_map_primordial _commuting	T1_energy_moment um_primordial_commuting
T1_current_primo rdial_commuting	T1_general_poten tial_primordial_ commuting	T1_scalar_and_ki netic_primordial _grassmann
T1_Lagrangian_pr imordial_grassmann	T1_Lagrangian_ne gative_controls_ primordial_grassmann	T1_field_equatio n_covariance_pri mordial_grassmann
T1_Euler_Lagrang e_derived_primor dial_grassmann	T1_Euler_Lagrang e_map_primordial _grassmann	T1_energy_moment um_primordial_grassmann
T1_current_primo rdial_grassmann	T1_scalar_and_ki netic_diagonal8_ commuting	T1_Lagrangian_di agonal8_commuting
T1_Lagrangian_ne gative_controls_ diagonal8_commuting	T1_field_equatio n_covariance_dia gonal8_commuting	T1_Euler_Lagrang e_derived_diagon al8_commuting
T1_Euler_Lagrang e_map_diagonal8_ commuting	T1_energy_moment um_diagonal8_commuting	T1_current_diago nal8_commuting
T1_scalar_and_ki netic_diagonal8_ grassmann	T1_Lagrangian_di agonal8_grassmann	T1_Lagrangian_ne gative_controls_ diagonal8_grassmann
T1_field_equatio n_covariance_dia gonal8_grassmann	T1_Euler_Lagrang e_derived_diagon al8_grassmann	T1_Euler_Lagrang e_map_diagonal8_ grassmann
T1_energy_moment um_diagonal8_grassmann	T1_current_diago nal8_grassmann	T1_scalar_and_ki netic_pointwise_ commuting
T1_Lagrangian_po intwise_commuting	T1_Lagrangian_ne gative_controls_ pointwise_commuting	T1_field_equatio n_covariance_poi ntwise_commuting
T1_energy_moment um_pointwise_commuting	T1_current_point wise_commuting	T1_scalar_and_ki netic_pointwise_ grassmann
T1_Lagrangian_po intwise_grassmann	T1_Lagrangian_ne gative_controls_ pointwise_grassmann	T1_field_equatio n_covariance_poi ntwise_grassmann
T1_energy_moment um_pointwise_grassmann	T1_current_point wise_grassmann	Gamma_properties

T1, sympy (23 checks):

check	check	check
T1.general_field .matrix_identities	T1.general_field .random_instance .commuting	T1.general_field .random_instance .grassmann
T1.metric.commut ing.S_invariant	T1.metric.commut ing.lagrangian	T1.metric.commut ing.negative_controls
T1.metric.commut ing.euler_lagran ge_derived	T1.metric.commut ing.euler_lagrange_map	T1.metric.commuting.emt
T1.metric.commut ing.pair_total_emt_zero	T1.metric.commut ing.current	T1.metric.grassm ann.S_invariant
T1.metric.grassm ann.lagrangian	T1.metric.grassm ann.negative_controls	T1.metric.grassm ann.euler_lagran ge_derived
T1.metric.grassm ann.euler_lagrange_map	T1.metric.grassmann.emt	T1.metric.grassm ann.pair_total_emt_zero
T1.metric.grassm ann.current	T1.metric.commut ing.general_potential	T1.metric.grassm ann.general_potential
T1.diagonal8.commuting	T1.diagonal8.grassmann	

T2, Wolfram (28 checks):

check	check	check
T2_Pn_in_Pin44	T2_Pn_covers_the _reflection	T2_character_of_Pn
T2_Gamma_times_P n_is_gamma_n	T2_kernels_gamma _n_with_frame_re flection	T2_frame_reflect ion_pointwise_x1 _commuting
T2_frame_reflect ion_pointwise_x1 _grassmann	T2_frame_reflect ion_pointwise_x2 _commuting	T2_frame_reflect ion_pointwise_x2 _grassmann
T2_frame_reflect ion_pointwise_x3 _commuting	T2_frame_reflect ion_pointwise_x3 _grassmann	T2_frame_reflect ion_pointwise_x4 _commuting
T2_frame_reflect ion_pointwise_x4 _grassmann	T2_frame_reflect ion_pointwise_x5 _commuting	T2_frame_reflect ion_pointwise_x5 _grassmann
T2_frame_reflect ion_pointwise_x6 _commuting	T2_frame_reflect ion_pointwise_x6 _grassmann	T2_frame_reflect ion_pointwise_x7 _commuting
T2_frame_reflect ion_pointwise_x7 _grassmann	T2_frame_reflect ion_pointwise_x0 _commuting	T2_frame_reflect ion_pointwise_x0 _grassmann
T2_mirror_is_isometry	T2_mirror_Lagran gian_commuting	T2_mirror_energy _momentum_and_cu rrent_commuting
T2_mirror_Euler_ Lagrange_commuting	T2_mirror_Lagran gian_grassmann	T2_mirror_energy _momentum_and_current_gr assmann
T2_mirror_Euler_ Lagrange_grassmann		

T2, sympy (16 checks):

check	check	check
geometry.brane_degenerate	geometry.mirror_isometry	T2.general_field.matrix_identities
T2.general_field.reflection_table	T2.metric.commuting.reflection_table	T2.metric.commuting.euler_lagrange_map
T2.metric.commuting.emt	T2.metric.commuting.current	T2.metric.commuting.S_odd
T2.metric.grassmann.reflection_table	T2.metric.grassmann.euler_lagrange_map	T2.metric.grassmann.emt
T2.metric.grassmann.current	T2.metric.grassmann.S_odd	T2.diagonal8.frame_reflections.commuting
T2.diagonal8.frame_reflections.grassmann		

Q, Wolfram (12 checks):

check	check	check
Q_Krein_metric_of_images	Q_symplectic_kernel_commuting	Q_generators_of_the_image_commuting
Q_symplectic_kernel_grassmann	Q_generators_of_the_image_grassmann	Q_one_particle_flat_dispersion
Q_one_particle_maps	Q_one_particle_Krein_signatures	Q_one_particle_Krein_inertia_real_frequencies
Q_one_particle_complex_and_zero_frequencies_Krein_neutral	Q_one_particle_general_field	Q_no_identification_of_independent_universes

Q, sympy (10 checks):

check	check	check
Q.canonical_anticommutator	Q.image_krein_metric	Q.image_own_quantisation
Q.image_generators_same_dynamics	Q.no_cancellation_independent_universes	Q.one_particle_maps
Q.one_particle_Krein_inertia	Q.one_particle_Krein_inertia_proof	Q.one_particle_complex_frequency_Krein_neutral
Q.T2_image_keeps_B		

Common checks (gammas, geometry, comparison with the Wolfram record), sympy (18 checks):

check	check	check
gammas.clifford	gammas.C	gammas.Gamma
gammas.Gamma_ant icommutes	gammas.B	gammas.equal_wol fram_fixture
geometry.omega_a ntisymmetric	geometry.gamma_c ovariantly_constant	geometry.spin_co nnection_components
compare.theory.status	compare.theory.r eflection_table	compare.theory.k rein_signs
compare.theory.o ne_particle	compare.theory.t heorem_T1	compare.theory.t heorem_T2
compare.theory.t heorem_Q	compare.theory.n ot_established	geometry.connect ion_depends_on_b4

Common checks (fixture, test fields, connections, determinism), Wolfram (11 checks):

check	check	check
fixture_Clifford _relation	fixture_definitions	C_and_B_basic
primordial_vielbein	pointwise_field_ is_general	connection_primordial
connection_mirror_patch	connection_diagonal8	connection_pointwise
theory_file_dete rministic	primordial_conne ction_depends_on_b4	

9.3 Re-verification for this document

For the first edition (2026-10-01) both pairing verifiers were re-run on a copy of the working tree: 99 of 99 and 64 of 64 checks passed and the outputs were byte-identical to the recorded files. After the review of the same day the pairing verifiers were extended (the Krein inertia of every real-frequency eigenspace, the Krein-neutral imaginary-frequency eigenspaces) and re-run on the working tree: `verify_pairing.wls` 101 of 101 PASS in about 80 s, `check_pairing.py` 66 of 66 PASS in about 140 s; the T3 verifiers 10 of 10 (about 3 s) and 13 of 13 (about 1 s). Two runs of each give byte-identical outputs. The completion records of section 8.5 (2026-10-08) are cited with the counts of their committed reports (Wolfram 3 of 3, sympy 7 of 7, the two demonstrations 7 of 7 each); their run times on the development machine, shared with other jobs, are about 15 s and 2 s for the completion verifiers, 10 to 15 min for the Rust demonstration (630 solver runs, 8 at a time) and about 10 min for the reference demonstration. The reports cited in this document are the files of the working tree. On 2026-10-14, after the relabel to the author's coordinates $x_0, \ldots, x_7$ (every run within 423 s), the pairing verifiers gave 101 of 101 (two runs with byte-identical outputs, 79 s and 91 s of kernel time) and 66 of 66 (194 s), the T3 verifiers 10 of 10 and 13 of 13, the completion verifiers 3 of 3 and 7 of 7, and the two demonstrations 7 of 7 each (the Rust demonstration with 10 workers in 281 s, the reference demonstration with 22 workers in 169 s); every report has the same checks with the same verdicts as before, apart from the two frame-reflection checks of the hidden direction, which are now named for x_0; the provenance files of the pairing record list the runs.

On 2026-10-16, after the move to the metric with $b_4(x_4)$ (every run within 423 s), the pairing verifiers gave 102 of 102 (two runs with byte-identical outputs, 112 s and 100 s of kernel time) and 67 of 67 (202 s; a second run of 177 s wrote a byte-identical report), the T3 verifiers 10 of 10

and 13 of 13, the completion verifiers 3 of 3 and 7 of 7 (each twice, byte-identical), and the two demonstrations 7 of 7 each (the Rust demonstration with 10 workers in 316 s and, from scratch, 294 s, byte-identical; the reference demonstration with 22 workers in 140 s and 131 s, byte-identical tables); the provenance files of the pairing record list the runs.

9.4 The metric with $b_4(x_4)$

On 2026-10-14 the author changed the primordial metric (section 2.1): the three extra-time components became

$$g_{55} = g_{66} = g_{77} = -e^{-2a_4(x_4)}\, b_4(x_4)\, \sin^{1/3}(6Hx_0),$$

and on 2026-10-15 the user ordered that everything be moved to $b_4(x_4)$, b_4 never 1; $b_4 = 1$ is the withdrawn metric for which this document was first written. The change and its first consequences are recorded in `Revision/metric_b4/METRIC_CHANGE_PROVENANCE.md`; since 2026-10-16 the pairing record itself is computed in the metric with b_4:

1. T1, T2 and Q: the pairing verifiers build the primordial and the mirror fields with the extra-time vielbein $e^{-a_4} b_4^{1/2} \sin^{1/6} z$ and $\sqrt{|g|} = b_4^{3/2} \cos z$, a_4 and b_4 arbitrary functions of x_4, and check that the spin connection contains b_4' (`primordial_connection_depends_on_b4`, `geometry.connection_depends_on_b4`), so every T1, T2 and Q check in these fields is made with b_4 not constant (102 of 102 and 67 of 67). The general-field checks (the linearity argument, the diagonal field with eight arbitrary functions, the pointwise field) cover every metric anyway. The Z2 mirror $x_0 \to \pi/(6H) - x_0$ is an isometry for every b_4, since b_4 depends on x_4 only (`T2_mirror_is_isometry`); the Z2 brane remains ASSUMED. The earlier argument of `Revision/metric_b4/results/consequences-b4.json` (4 of 4 PASS) reached the same conclusions from the factorisation of the operators.

2. T3: the Kohn-Sham record of 2026-10-16 is that of the metric with b_4: the ansatz $\Psi = e^{ik\cdot x} W^{-3} b_4^{-3/4} \chi$ removes the b_4' term of the spin connection and b_4 does not enter the block Hamiltonians h_j (`ansatz_removes_spin_connection`, `ansatz_without_b4_factor_term_survives` of both Kohn-Sham theory reports), so the proof of section 8.3 holds slice by slice for every b_4; the theorem record names the slice $b_{4,0}$ in (H3.1) and (H3.6), its checks are unchanged (10 of 10, 13 of 13).

3. The Kohn-Sham source verdict of section 8.4 is that of the metric with b_4 (`Revision/field_equations_a4/reports/ks-source-conditions.json`, 6 of 6 PASS: C1 for every a_4 and b_4, C2 and C3 on the history, and the off-diagonal condition `ks_offdiagonal_condition_on_history`, which holds), and the field equations for a_4 and b_4 are verified in `Revision/field_equations_a4/reports/wolfram-a4-report.json` (59 of 59 PASS) and `Revision/field_equations_a4/reports/python-a4-report.json` (65 of 65 PASS). Corollary C1 (section 7) is stated for the metric with b_4.

4. The numerical demonstrations of sections 8.4 and 8.5 were recomputed on the history with b_4 of the Kohn-Sham record (2026-10-16).

The answer of section 11.3 is unchanged. The change of the metric adds no creation process: nothing about the creation of universes, in pairs or otherwise, is proved with b_4 either.

10 Reproduction

From the repository root (the sympy pairing checker reads the Wolfram theory record for its final comparison, so the Wolfram verifier runs first; the Kohn-Sham theory checker reads the record written by the Kohn-Sham source-conditions checker, so that one runs before it, and so does the analysis of the a_4 equations with the Kohn-Sham source, which re-derives that record; the T3 completion runs after the T3 verifiers: the Wolfram completion verifier first, then the Rust demonstration (it needs the built Rust solver), then the reference demonstration (it compares its $-m$ universe with the Rust table), and the sympy completion checker last (it reads the completion record and both demonstration reports); `<target>` is the cargo target folder and `<scratch>` an empty working folder; the PDF command is one command, the backslash is the line continuation of a POSIX shell):

```
wolframscript -file Revision/algebra/wolfram/verify_algebra.wls
python Revision/algebra/python/check_algebra.py
wolframscript -file Revision/theory/wolfram/verify_field_theory.wls
python Revision/theory/python/check_field_theory.py
wolframscript -file Revision/theory/wolfram/verify_scope.wls
python Revision/theory/python/check_scope.py
wolframscript -file Revision/pairing/wolfram/verify_pairing.wls
python Revision/pairing/python/check_pairing.py
wolframscript -file Revision/field_equations_a4/wolfram/verify_field_equations_a4.wls
python Revision/field_equations_a4/python/check_field_equations_a4.py
wolframscript -file Revision/kohn_sham/theory/verify_ks_theory.wls
python Revision/field_equations_a4/python/check_ks_source_conditions.py
python Revision/field_equations_a4/ks_source/ks_source_a4.py
python Revision/kohn_sham/theory/check_ks_theory.py
wolframscript -file Revision/pairing/kohn_sham/wolfram/verify_t3.wls
python Revision/pairing/kohn_sham/python/check_t3.py
wolframscript -file Revision/pairing/kohn_sham/wolfram/verify_t3_completion.wls
cargo build --release --manifest-path Revision/kohn_sham/solver/Cargo.toml
python Revision/pairing/kohn_sham/numerics/t3_rust_demo.py \
    --solver <target>/release/revision_ks_solver --work <scratch> --jobs 10
python Revision/pairing/kohn_sham/numerics/t3_reference_demo.py --jobs 22
python Revision/pairing/kohn_sham/python/check_t3_completion.py
python scripts/build_provenance_pdf.py Revision/docs/PAIR_CREATION_PROOFS.md \
    --developer-layout --specifications Revision/pdf-specifications.json
python -m unittest Revision/tests/test_pair_creation_proofs_publication.py -v
```

The PDF build runs the Markdown-to-LaTeX builder twice (with different hash seeds), runs pdflatex three times into each of two fresh directories, requires warning-free logs and byte-identical PDFs, and checks the PDF against its entry `pair-creation-proofs` in `Revision/pdf-specifications.json` (the Revision registry; the registry of the earlier stages is not touched).

11 What is proved and what is not

11.1 Proved

Exactly, for both fields (dirac16complex with Grassmann components, dirac16complex00 with commuting components) unless stated, under the hypotheses stated with each result:

1. T1 (section 4): $\mathcal{L}_{m,\lambda}[\Gamma\Psi] = -\mathcal{L}_{-m,-\lambda}[\Psi]$; the solutions of the (m,λ) theory and of the $(-m,-\lambda)$ theory correspond one to one; $T \to -T$ and $J \to -J$; the pair has zero total energy-momentum, current and charge as classical bilinears; in every gravitational field taken as a fixed background.

2. T2 (section 5): Γ combined with a Pin(4,4) reflection of character -1 gives $\mathcal{L}_{m,\lambda}[\gamma^n\Psi; R_n e] = +\mathcal{L}_{-m,\lambda}[\Psi; e]$; in the author's field the mirror across the Z2 brane $z = \pi/2$ (ASSUMED construction) pairs a $(-m,\lambda)$ solution on the patch with an (m,λ) solution on the mirror patch at EQUAL energy-momentum and charge.

3. Q (section 6, dirac16complex only): the chirality image carries the Krein metric $-B$ and is the same quantum system re-labelled; an independently quantised $(-m,-\lambda)$ universe carries $+B$ and cannot be identified with it; the generators of two independent universes add without cancelling; the $+m$ and $-m$ one-particle ($\lambda = 0$) spectra are identical (flat space, or frozen coefficients at a point); the T2 image keeps $+B$.

4. C1 (section 7): a T1 pair as the complete classical source of the author's metric is a zero source; in Einstein gravity the author's metric then has no solution for $H > 0$, and with the author's weights no exponential history with b_4 not constant is a vacuum solution.

5. T3 (section 8, dirac16complex): every self-consistent instantaneous Kohn-Sham state with (m,λ,θ) is mapped by the block map of Γ, with the brane parities exchanged and $\theta \to \pi - \theta$, onto a self-consistent Kohn-Sham state with $(-m,+\lambda,\pi-\theta)$ with equal levels, occupations, Kohn-Sham energy and energy-momentum profiles, and $S \to -S$ (ASSUMED Z2 brane); with the completion (section 8.5), at the 16-component level with the expectation rule of the canonical quantisation, every component of the energy-momentum tensor and of the current is unchanged and S, Q change sign (S6), and the filling convention of the $+m$ member is carried onto the $-m$ member.

11.2 Not established

1. No creation process: the theorems map solutions to solutions and quantities to quantities; nothing in these equations produces a universe, a pair of universes or a change of the number of universes, and no transition between "no universe" and "two universes", no initial state, no vacuum decay and no tunnelling process is derived.

2. No rate, probability or amplitude: no transition amplitude, probability, cross-section, rate or Bogoliubov coefficient for creating universes of masses $\{+m,-m\}$ is computed or implied; no wave function of the universe and no path integral is part of these theorems.

3. No dynamical necessity: no equation and no conservation law forces the partner to exist; a single universe with mass $+m$ is an equally valid solution without its partner. T1 and T2 are correspondences between solutions of two parameter sets, not a mechanism.

4. T1 is not a symmetry of a single theory: it changes the parameters and the sign of the action, and for $\lambda \neq 0$ it pairs (m, λ) with $(-m, -\lambda)$, not $+m$ with $-m$ at fixed coupling. The pure pairing $(m, \lambda) \to (-m, \lambda)$ is T2, at equal, not opposite, energy-momentum: T2 does not reverse the sign of T.

5. The vanishing total energy-momentum and charge of a T1 pair holds for classical bilinears and as an operator identity within ONE quantum system; it does not hold for two independently quantised universes, whose generators add without cancelling.

6. Test-field statement: the gravitational field is fixed and the same for both members. The back-reaction through the field equations for a_4 and b_4 is not part of the theorems; corollary C1 is the only statement about it, and it concerns the sum of classical sources in one common geometry.

7. The Z2 brane: the mirror across $z = \pi/2$ uses the ASSUMED Z2 construction; the metric is degenerate there ($g_{00} = 0$ and $\sqrt{|g|} = 0$), and no junction condition, brane tension or matching of the field across the brane is derived.

8. Quantum positivity: in flat 4+4 space every real-frequency eigenspace of the one-particle Hamiltonian has Krein inertia (4,4) (proved, section 6.3), and the eigenspaces of imaginary or zero frequency are Krein-neutral; a positive-norm Fock space for either universe is not established by these theorems (the good-sector construction of the theory record is a separate result, made only for single good-sector momenta with frozen coefficients and not used here; section 6.3).

9. dirac16complex00 is a classical field: no quantum statement is made for it.

10. The Kohn-Sham level T3 holds for the instantaneous (adiabatic) mean-field Kohn-Sham states only, with the ASSUMED Z2 brane and the transformed tip condition; the time-dependent Kohn-Sham problem is open, no correlation is included, and T3 says nothing about back-reaction (section 8); the history $a_4 = AHx_4$, $b_4 = b_4(0)e^{(4/3)AHx_4}$ of the numerical demonstrations is a prescribed background, the filling of the Kohn-Sham states is a CONVENTION, and the demonstrations of section 8.5 are numerical, not proofs.

11.3 The answer to the request

The request "PROVE that Universes of masses {+mass, -mass} are created in pairs" is answered exactly as follows, for dirac16complex and for dirac16complex00. Proved: to every solution with mass m the explicit maps Γ (T1: with $\lambda \to -\lambda$, opposite energy-momentum and charge, in every gravitational field) and γ^0 with the ASSUMED Z2 mirror construction (T2: same λ, equal energy-momentum, in the author's primordial gravitational field) assign a solution with mass $-m$; for dirac16complex, at the Kohn-Sham level, the block map of Γ with the transformed boundary conditions assigns to every self-consistent instantaneous Kohn-Sham state with (m, λ) one with $(-m, +\lambda)$ at equal energies (T3, ASSUMED Z2 brane). That universes occur in pairs is not proved: these are maps between the solutions of two parameter sets, and a single universe with mass $+m$ is an equally valid solution without its partner. At the quantum level of dirac16complex the chirality partner is the same quantum system with the Krein metric $-B$, and two independently quantised universes of masses $+m$ and $-m$ have identical one-particle ($\lambda = 0$) spectra (flat space, or frozen coefficients at a point) and no cancelling generators. Not proved: that such universes are CREATED, in pairs or otherwise. No creation process, rate or amplitude follows from these equations.